This was supported by a successful vaccination campaign, a series of measures adopted to respond to the multifaceted nature of the shock, an extraordinary agricultural season, solid manufacturing and agro-industrial export, supportive counter-cyclical fiscal and monetary measures, and an unprecedented level of remittances.

In response to the drought, in 2022 the Government of Morocco (GoM) adopted a US$1 billion emergency recovery package for farmers, including agricultural insurance and rescheduling debt mechanisms.

Morocco·s New Development Model (NDM) presents a new vision for the country·s development. '(The structural reforms launched two decades ago gave way to a sustained period of economic growth and poverty reduction that is unparalleled in the country·s contemporary history.

Building a sustainable Morocco will require preserving natural resources, enhancing the resilience to climate change, and safeguarding water resources through more efficient use and management of its scarcity.

In his speech, His Majesty King Mohammed VI outlined four main guiding principles for water resources management in the context of increasing water deficits: (a) launching more ambitious programs and initiatives, and leveraging modern technology, particularly for water savings and wastewater reuse; (b) improving the rational exploitation and preservation of groundwater resources, in particular by curtailing illegal pumping and well digging; (c) fostering cross-sectoral coordination and updating sectoral strategies on a continuous basis, in light of their pressure on water resources and the need to build a water secured Morocco; and, (d) considering the real cost of water, accounting for each stage of the water mobilization process, and promoting transparency and awareness about all the drivers of cost of water.

According to the Country Climate and Development Report for Morocco (CCDR),[12] the greatest potential for putting Morocco on a climate-resilient and low-carbon pathway lies in (a) tackling water scarcity and droughts, notably through the nexus of water and agriculture; (b) enhancing resilience to floods to preserve urban and coastal economies and livelihoods; and (c) decarbonizing the economy for a zero-net emission pathway by the 2050s. The CCDR highlights the need for improved inter-institutional coordination[13] and mainstreaming climate actions into economic, fiscal, and financial policies.

[13] A rapid Climate Change Institutional Assessment (CCIA) was conducted as part of the preparation of the CCDR.

The water deficit between supply and demand in 2020 was estimated at 1.8 billion cubic meters (BCM) per year, with demand expected to increase by 15 percent between 2020 and 2050.[14] Water security challenges are compounded by the unequal distribution of water over space and time, with most of the surface water resources concentrated in the northwest of the country.[15] Surface water shortages have led to the over-exploitation of groundwater resources, which are being abstracted faster than they are being replenished.

Water shortages are further aggravated by the loss of existing storage due to high dam sedimentation rates,[17] water pollution, and saline intrusion in the coastal regions.

Hotter and drier conditions are also expected to increase demand, with crop water requirements predicted to increase by up to 12 percent, increasing demand for irrigation and placing further demands on limited water resources. These changes in supply and demand could result in a deficit of between 4 to 7 BCM per year by 2050.[19] Such constraints could significantly affect the Moroccan economy,[20] with water scarcity exacerbating competition among water users and potential causing a 6.5 percent decline in real GDP under extreme conditions.[21]

8. To address the country·s water challenges, the GoM is implementing comprehensive large-scale water sector investment programs. These include the National Program for Potable Water Supply and Irrigation (Programme National pour l·Approvisionnement en Eau Potable et l·Irrigation; PNAEPI) 2020-2027, adopted in January 2020, which encompasses the first phase of the draft National Water Plan (Plan National de l'Eau, PNE) and outlines an ambitious set of investments. This builds on a long tradition of water resource development, with 135 large dams providing 17.5

[14] Total water demand by 2050 is expected to be between 18.7 and 20 BCM per year taking climate change into account. Water demand from uABHn households is expected to increase by 50 percent (from 1.1 to 1.7 BCM per year) during the same period, with corresponding increases in effluent quantities.

Source: Ministry of Equipment, Transport, Logistics, and Water: Draft National Water Plan 2020·2050.

[15] Ministry of Equipment, Transport, Logistics, and Water: Draft National Water Plan 2020·2050.

Source: Ministry of Equipment, Transport, Logistics, and Water: Draft National Water Plan 2020·2050.

Source: Ministry of Equipment, Transport, Logistics, and Water: Draft National Water Plan 2020·2050.

However, there is also recognition that increasing resilience to climate change requires ambitious policy reforms that reflect the value of water, increase the transparency of costs along the water value chain, balance competing demands, and incentivize more efficient and rational uses The PNAEPI acknowledges the incremental cost of new water supplies and the need for strong policy reforms, establishing targets for reducing water-losses, including provisions for the reduction of dam sedimentation and reforestation of watersheds, incentives for water savings in agriculture, drinking water, tourism, and industry, along with the preservation of groundwater resources. The PNAEPI is accompanied by a communication strategy, adopted in 2021, to raise awareness and change behaviors relating to water conservation and its value.

The governance of the water sector is evolving in response to the changing nature of water resources and the new socio-economic vision. The legal and institutional system integrates religious tradition and customs alongside modern provisions introduced since independence in 1956.[2] 3 The Law 10-95 (1995) stipulated the rules for integrated, decentralized, and participatory water resources management to guarantee the right of Moroccans to water, including the creation of River Basin Agencies (Agences du Bassin Hydraulique, ABHs) as technical and financially independent entities responsible for planning and managing water resources at the basin level. [2] 4 The Water Law 36-15 (2016) retained and re-enforced the foundational elements aimed at improving water use efficiency, providing universal access, reducing disparities between rural and urban areas, and ensuring water security. It further emphasized decentralized, integrated, and participatory management and planning of water resources; strengthened consultation and coordination by establishing river basin councils and the adoption of the PNE by the Superior Council of Water and Climate; established the legal foundations to diversify sources of supply through the use of unconventional water resources; mandated the establishment of water information systems at national and river basin levels; and strengthened mechanisms for the protection and conservation of water resources (raw water abstraction and pollution fees and participatory groundwater management contracts). As part of the continued commitment to régionalisation avancée, the GoM plans to establish regional multiservice companies (Sociétés Régionales Multiservices, SRM) for energy and potable water distribution and wastewater collection and treatment, to substitute autonomous municipal public utilities (Régies).[2] 5

10.

[2] 4 Their duties include preparing river basin management plans, authorizing water abstractions and discharges, and maintaining a public register; collecting fees for abstraction and effluent discharges; providing financing and technical assistance for water pollution prevention; ensuring efficient water use; monitoring water quality; enforcing laws related to water resource protection; setting up an emergency response system; and promoting public awareness about water resource management.

The ongoing transformations in Morocco·s water sector have significant implications for its financial sustainability. The upscaling of non-conventional water resources has important implications for the country·s water balance and the associated cost structure for the sector. Production and retail water supply tariffs and sanitation fees do not fully reflect production and distribution costs and are generally insufficient to cover operational and maintenance costs. Financial sustainability is further undermined by the under-collection of raw water abstraction and water pollution fees.[28] The lack of a robust financing strategy for the sector could present barriers to the realization of the GoM·s ambitious plan for expanding desalination capacity, particularly for irrigation purposes, and compromise the sector·s financial viability with impacts on future water security. Even if technological advances continue to reduce the cost of desalination,[29] capital, and operating costs would remain significant and energy-dependent, potentially exacerbating the sector·s financial fragility. The evolution of the sector illustrates the challenges in positioning water in the political-administrative ecosystem and, simultaneously, adapting to its complex interrelation between different sectors,[30] and the uncertainty imposed by climate change. In this context, an enhanced governance model is needed to ensure an integrated approach to water production and distribution, optimize allocation decisions, ensure financial sustainability, and ensure that decisions about water are made to maximize socio-economic outcomes in alignment with the development model for Morocco.

Specifically, the Program supports Focus Area 3 ·Promoting Inclusive and Resilient Territorial Development·, directly contributing to the three objectives of: (a) improving the performance of key infrastructure delivery services to cities; (b) improving access to sustainable water resources; and, (c) enhancing adaptation to climate change and resilience to natural disasters.

In addition to technological advances and economies of scale achieved by larger plants, falling costs are driven by project development choices, such as colocation of desalination plants with power plants, and enha competitiveness from more efficient methods of project financing and delive

The Program further contributes to the four pillars of the WBGs Global Crisis Response Framework[37] that is aimed at helping countries navigate multiple, compounding crises by strengthening resilience and supporting paths to build long-term resilience, while strengthening policies, institutions, and investments.

The Program builds on the GoM·s experience with the PforR instrument and the Government·s successful experience with results-based financing in various sectors, ranging from agriculture to education, early child development and urban development, among others. The Program focuses on a portion of the PNAEP where the Government aims at enhancing efficiency, effectiveness, and impact of expenditures by linking the disbursement of funds to the achievement of specific results based on the existing performance-based budget system.

The Program will provide a platform for national and local entities to incentivize sustained progress of the government reform program, including on water valuation, state-owned enterprises, and the régionalisation avancée.

The Government program is defined by the National Program for Water Supply and Irrigation (2020-2027 PNAEPI: Programme National pour l'Approvisionnement en Eau Potable et l'Irrigation). Based on the orientations provided by His Majesty King Mohamed VI, in 2017, a technical commission - under the guidance of the Interministerial Water Commission - was charged with the preparation of a program to accelerate investments in the water sector to strengthen drinking water supply and irrigation, in particular for the areas most impacted by the successive droughts impacting the country since 2015.[38] With investments estimated at MAD 143 billion (US$ 14.4 billion), the PNAEPI is expected to be financed through a combination of public subsidies, user contributions, and a combination of the private sector, climate financing, and international financing institutions.

World Bank Group Climate Change Action Plan 2021·2025: Supporting Green, Resilient, and Inclusive Development.

[37] World Bank (2022) Navigating Multiple Crises, Staying the Course on Long-term Development: The World Bank Group·s Response to the Crises Affecting Developing Countries.

[38] The technical commission composed of representatives of the Ministry of the Interior, the Ministry of Agriculture, Maritime Fisheries, Rural Development, Water and Forests, the (then) Ministry of Economy, Finance and the Reform of the Administration, the (then) Ministry of Equipment, Transport, Logistics and Water, the (then) Ministry of Energy, Mines and the Environment, and the National Office for Electricity and Water Potable (ONEE).

The PNAEPI is the first tranche of the draft PNE, which is under preparation and shall be adopted by Decree after the review by the Superior Council of Water and Climate (chaired by the Prime Minister).

The PNAEPI aims to diversify the sources of supply, guarantee water security, and reduce climate change impacts by accelerating investments to strengthen water supply for drinking and irrigation uses. The PNAEPI's major objectives are: (a) expanding water supply by increasing water storage capacity and the contribution of non-conventional sources to the water matrix (wastewater reuse and desalination), and safeguarding groundwater resources; (b) improving water efficiency by reducing water losses in conveyance and distribution networks (potable water and irrigation) and improving water productivity in the irrigated agriculture sector; (c) strengthening access to potable water supply in rural areas; and (d) increasing awareness on the value of water. Among other factors, the successful implementation of the PNAEPI relies on: (a) boosting cross-sectoral coordination; (b) re-thinking the sector financial framework to reflect the increasing cost of water mobilization; (c) adopting critical regulations relating to non-conventional water resources, participative aquifer management contracts, and water data management and information systems; (d) strengthening the performance of ABHs, in particular concerning water withdrawal control and enforcement; and (e) supporting the on-going reform of the potable water and wastewater service sector aimed at improving service delivery and financial sustainability.

The institutional arrangements for implementing the PNAEPI reflect water·s cross-sectoral nature and the GoM·s high-level commitment towards water security.

The Inter-Ministerial Technical Committee (chaired by the Minister of the MEE) ensures the PNAEPI·s coordination and implementation consistency, maintains a dashboard on the country·s water supply situation and program implementation, proposes adjustments to the PNAEPI; and defines the annual communication program. Regional Committees (chaired by the Wali of the region) define regional needs related to small dams and drinking water supply in rural areas and monitor program implementation at the territorial level. In addition to the concerned institutions (MEF, MEW, MoI, MAF, MEM, and ONEE), activities under the PNAEPI are implemented by several institutions, including the ABHs, water supply operators (ONEE, Régies, private concessionaires), Regional Agricultural Development Offices (Office Régional de Mise en Valeu Agricole, ORMVAs), and the OCP Group

The Theory of Change outlines the Program logic and its contribution to the water-related development challenges in Morocco.

The Program will support interventions that address the water challenges on three levels: (a) sector-level governance focused on water planning, strengthening of groundwater management, basin-level capacity to manage water resources, data and information management systems; and performance of water and wastewater operators; (b) improvement of the financing sustainability of the sector, including through local-level improvements to improve water use efficiency and reduce non-revenue water; and, (c) promote non-conventional water, from the regulatory perspectives and by supporting investment to scale-up wastewater reuse.

The scope and boundaries of the Program are based on the activities identified in the PNAEPI and defined by priorities that contribute to the following: (a) strengthening the governance of the water sector; (b) improving the valuation and financial sustainability of the water sector; (c) creating a water-savings culture and reducing water losses; and (d) providing an enabling environment for the integration of non-conventional water resources.

The selection also takes into consideration the increasing reliance of these areas on desalinization with all of these ABHs, except for Loukkos and Sebou, experiencing extreme water shortages. The ABHs will be responsible for targeted activities within their geographic responsibility (including river basin data and information systems, performance improvement, installation of smart meters for groundwater withdrawals· monitoring, and aquifer management contracts). Program related activities within the Régies are also located within the geographical area of these ABHs, including activities such as water loss management and wastewater reuse projects, contributing to the overall water security within these areas.

These include support for the GoM·s efforts to adapt the framework governing the sector to ensure that it is fit for purpose and deliver on Morocco·s expectations related to achieving longer-term goals of water security, recognizing the value of water, and ensuring the integration of non-conventional sources of water. This approach is built on: (a) strengthening the water sector governance; (b) improving financial sustainability and water use efficiency, and (c) enabling the integration of non-conventional water resource

RA 1: Strengthening Water Sector Governance aims to strengthen the sector·s governance for sustained water resource management. Activities include: (a) the preparation and adoption of the PNE as the Borrower·s framework based on a long-term strategy that reflects the increasing uncertainty due to climate change exacerbated vulnerabilities such as increased water scarcity and drought, and including the definition of principles of water valuation considering primarily economic and financial aspects of water; (b) the development, adoption, and implementation of regulatory instruments and consultative processes to improve implementation of participative aquifer management contracts; (c) the preparation and signature of participative aquifer management contracts in three selected aquifers; (d) the installation of smart meters for measuring groundwater withdrawals from large users and reducing withdrawals or demand thus increasing water availability during drier months; (e) the development, implementation, and adoption of a performance benchmarking framework to strengthen the performance of selected ABHs to deliver on their core functions of planning, managing, developing and protecting water resources, and operating and maintaining of infrastructure; (f) the operationalization of water information systems (SNIEAU and river basin); (g) improvements in water data management and information management systems, including regulations, formal specifications, and benchmarking for data generation, sharing, and access, quality assurance and control standards, upgrade, equipment, and maintenance of monitoring and information systems; (h) the installation and rehabilitation of hydrological stations and piezometers (integrating climate resilient design); and, (i) the operationalization of multiservice operators (energy and water supply distribution, and wastewater collection and treatment) performance systems and adoption of minimum service standards.

RA 2: Improving Financial Sustainability and Water Use Efficiency aims to enhance climate resilience by improving water valuation, reduce water losses from existing distribution systems, and encourage water conservation through communication campaigns. Activities include: (a) the development of a Financial Sustainability Framework for the sector, including the development of a financial model for the sector to inform pricing strategies for specific sub-sectors and the adoption of a financial sustainability action; (b) the implementation of the PNAEPI communication strategy to raise awareness of the importance of water conservation, including baseline and end-strategy impact evaluations; and, (c) the implementation of water loss reduction plans, including the deployment of geographical information management systems (GIS) and hydraulic models; meters deployment (bulk- and micro-meters), network sectorization and pressure control program; and, leakage detection and rehabilitation campaigns.

RA 3: Enabling the Integration of Non-Conventional Water Resources aims to improve the enabling environment for non-conventional water resources and scale-up of treated wastewater reuse.

Total program financing over the World Bank fiscal years 2023 - 2028 is expected to be MAD 5,801 million (equivalent to US$ 573 million). Of this, an expected US$ 223 million equivalent (38.6 percent) will be funded by the GoM, and US$ 350 million equivalent (61.4 percent) will be financed through an IBRD loan.

The PDO is to strengthen water sector institutions and increase water availability in selected areas in Morocco.

Water information systems operationalized (value).
Potable water savings in distribution water supply networks (m3).

The PDO-level indicators were selected to measure the key achievements that the Program should reach within the implementation period and promote institutionalization and sustainability of the Government·s efforts under the PNAEPI.

This DLI aims to incentivize the adoption of the PNE, which will set the 30-year vision for the water sector.

The preparation of the Plan entails significant coordination efforts among the relevant stakeholders, including the MEF, MEE, MoI, MAPMDREF, OCP Group, and service providers (irrigation and water supply and sanitation services); and reflects the water needs of sector-specific plans (notably the Generation Green Plan and OCP Group Water Plan). The Superior Council of Water and Climate, chaired by the Prime Minister, will validate the PNE and it will be adopted by decree of MEE.

Finally, the DLI supports GoM's effort for improved groundwater management by deploying smart meters to monitor groundwater abstraction in aquifers within the Program area, which will simultaneously increase water savings and reduce GHG emissions.

This DLI seeks to incentivize improved content, quality, accessibility, and use of water data.

Finally, the DLI provides incentives for publishing the status of the annual hydrological situation, including water uses and water quality within the jurisdiction of the ABHSs included in the Program.

These tools will facilitate GoM decision-making to bridg the gap between cost and revenues along the water value chain (including wat mobilization through wastewater reuse and disposal), particularly considering significant transformation of the water matrix envisaged under the PNAE In particular, the DLI will support developing a financial model and adopting a financial sustainability action plan. All these tools will consider relevant costs (CAPEX and OPEX) and revenues (tariffs, transfers, and taxes).

This DLI seeks to incentivize demand management efforts in the distribution systems by implementing non-revenue water reduction plans by Régies in the Program area.

This DLI supports the GoM's ambition to scale up treated wastewater for productive uses (green spaces, golf, industry, tourism, agriculture, and others) as a strategic resource to mitigate increasing water scarcity risks. Through this DLI, the Program will support the adoption of the bylaw defining the norms for wastewater reuse for agricultural purposes, noting that the general use of wastewater is already adopted. In addition, the DLIs will incentivize the implementation of treated wastewater re projects, including expanding the installed capacity to the treatment level required reuse and the distribution network from the wastewater treatment plant to th taker of the treated effluen The Program will make 52 million m3 of treated wastewater available for reuse, thus enhancing resilience by reusing treated wastewater and increasing availability of freshwater resources for alternate uses (equivalent to 52 percent of the PNAEPI target).

The governance measures are intended to address the challenges of climate change-exacerbated water scarcity and the increased uncertainty in future water variability through incentivizing the incorporation and mainstreaming of measures in the data, information, and planning processes. This includes specific provisions to incorporate climate change models and considerations into the PNE, and the performance evaluation framework for the ABHs. In addition to the contribution to the adaptation agenda, the Program includes several mitigation measures, specifically through the reduction in energy consumption as a result of reduced water losses that will generate savings in GHG emissions.

The PCU, nested at the MEE, will coordinate the timely consolidation and provision of monitoring data and supporting documents to the IGF and the World Bank.

The respective general inspection (See Annex 2) will be responsible for reviewing and approving the reports prepared by the PCU and submitting the verification results, along with any supporting documentation, to the MEF for subsequent transmission to the World Bank.

These are designed to ensure joint ownership by all stakeholders and reflect the multi-stakeholder and partnership-based nature of the PNAEPI, which relies on a close collaboration between actors across local, regional, and national tiers.

The Technical Committee is required to meet as necessary, but at least twice a year, and is responsible for monitoring implementation of the PNAEPI. Specific responsibilities include: (a) ensuring PNAEPI·s coordination and implementation consistency; (b) establishing and informing a dashboard about the country's water resources and the progress of the implementation of the program; (c) proposing adjustments to the PNAEPI·s based on the water resources situation and the results of the studies carried out; and (d) overseeing the annual communication program.

The MEE will be responsible for overall Program management through a Program Coordination Unit (PCU) established in the DRPE. The PCU will be responsible for coordinating overall implementation, consolidating semi-annual progress reports submitted by the implementing entities, monitoring and reporting of DLIs, financial management and audits, and ensuring stakeholder coordination.

The POM will define the staffing requirements for the core team within the PCU, including, at a minimum, a Program Director, financial, procurement, environmental and social (E&S) specialists.

The PCU will be supported throughout implementation by technical assistance to be funded through the corresponding government budget lines.

The implementing entities will be responsible for financial management, procurement quality control and management, compliance with E&S requirements, and reporting as per the POM. The implementing entities will be supported by specialized technical assistance, which will be reflected in the respective budget lines of the implementing entities.

1	National Water Plan (PNE) Adopted	MEE
2	Groundwater management improved	MEE and ABHs included in the Program
3	ABH performance framework adopted and ABH performance improved	MEE and ABHs included in the Program
4	Water Information Systems operationalized and used for decision-making	MEE and ABHs included in the Program
5	Operators' performance information system operationalized	MoI and Régies included in the Program
6	Financial sustainability framework of the water sector improved	MEF
7	Volume of potable water savings in distribution water supply networks	Régies included in the Program
8	Wastewater reuse scaled-up	MoI and Régies included in the Program

38. The World Bank will provide implementation support based on the detailed Implementation Support Plan (Annex 7) the focus of which will be on the timely implementation of the agreed Program Action Plan (PAP) (Annex 6), provision of necessary technical support, conducting of fiduciary reviews, and monitoring and evaluation activities.

The POM will provide overall guidance on Program implementation and will include information such as the detailed project description, financing plan, roles, and responsibilities for implementing agencies, the role of the World Bank during implementations, PCU staffing requirement, fiduciary requirements (procurement, financial management and fraud and corruption), E&S technical manual, and M&E arrangements.

The PCU will be responsible for consolidating all the data and documentation necessary for monitoring, verification, and evaluation purposes. The PCU will bear the responsibility for monitoring overall progress toward achievement of the Program's results and ensuring timely collection and provision of monitoring data and verification documents for the World Bank and IVA.

Once the World Bank has notified the Government in writing that it has accepted evidence of the achievement of the indicator and the amount of the eligible payment, the MEF will then submit applications for withdrawal per the amounts allocated to individual DLIs. The withdrawal amount against the DLIs achieved will not exceed the amount of the financing allocated by the World Bank for the specific indicator.

The Program-financed results areas are embedded in the budget and expenditure management processes of the country system. All payments of the Program will be made through the centralized Treasury Bank accounts held at the Central Bank (Bank Al-Maghrib). The GoM, through its budget, will transfer the funds to the MEE and MoI based on the expenditure framework and activities to be executed by the directorates and agencies involved in the Program and thus prefinance the expenditure.

For advances and achieved results, the funds will be disbursed to the GoM·s Treasury Single Account opened at the Bank Al-Maghrib.

The Program includes capacity-building activities to strengthen technical, E&S, and fiduciary aspects. Focal points of participating entities will be equipped and trained on identifying E&S risks and impacts regarding national regulations and international practices; the definition, implementation, and monitoring of mitigation measures; and the reporting and evaluation of the Environmental and Social Management System (ESMS). Fiduciary capacity building will include (a) monitoring fiduciary implementation progress; (b) supporting the Government to resolve implementation issues and carry out institutional capacity building; and, (c) complying with audit reports.

A Bank-executed Trust Fund (US$ 600,000) from the Global Facility for Disaster Reduction and Recovery (GFDRR) was secured to support the RA 1, specifically advising on the optimization of existing monitoring networks and issues around dams· sedimentation Also, for the same RA, Bank-executed Trust Funds have been secure (US$ 400,000) from the Global Water Security and Sanitation Partnership (GWSP), specifically in guiding international experience and appropriate options to inform the development of institutional mechanisms for the water sector in Morocco, including guiding the development of a performance benchmarking framework for ABHs. Similarly, resources from GWSP will support the achievement of results under RA 2 on Improved Financial Sustainability and Water Use Efficiency through international experience and approaches to the development of financial models and strategies to improve the sustainability of the sector, and international experience in formulating campaigns to influence water conservation behavior.

Capacity building and institutional strengthening will leverage synergies with other activities and Programs supported by the World Bank.

This will help inform the preparation of the financing strategies for selected sub-sectors under RA2. Also, efforts under the Program will be closely coordinated with efforts made by the General Meteorological Direction (Direction General de la Météorologie) and the National Agency on Water and Forest (Agence Nationale des Eaux et Forêts) under the Morocco Climate Operation / Support to Morocco·s Nationally Determined Contribution PforR (P178763; approved by the Board in June, 2023).

The Program·s strategic relevance is further underscored by its contribution to Morocco·s NDC for climate change mitigation and adaptation. The Program will support the country·s climate adaptation priorities of reducing the impacts of water scarcity, droughts, and floods. This will be achieved by strengthening the institutional framework for managing the development of water resources under increasing uncertainty due to climate change through the preparation and adoption of the PNE, enabling the introduction of non-conventional water resources, strengthening data and knowledge to inform more robust investment planning and decision-support systems, including contingency preparedness and responses to extreme hydroclimatic events, and improving the understanding and management of strategic groundwater resources. The Program will also directly support the emissions reduction target in the NDC through investments in wastewater treatment and the energy efficiency gains that can be realized through investments in water loss reduction in distribution networks.

This facilitated the exclusion of high-risk activities -such as large-scale civil works related to desalination- and procurement of works estimated to exceed the Bank·s thresholds. There is a high degree of predictability in the timely release of budget appropriations, and the sustainability of the expenditure framework of participating entities is assured.

The Finance Laws (Lois de finances) for 2021, 2022, and 2023 include specific provisions for the implementation of measures envisaged under the PNAEPI aimed at increasing water availability, such as projects related to water transfers, seawater desalination, and the reuse of treated wastewater. As a result, the alignment of the budget with government priorities, classification, sustainability, and predictability is considered sufficient.

The financial sustainability of the Program is assured by the Government·s overall medium-term budget trajectory.

An evaluation of the costs and benefits confirms the economic rationale for the Program.4 [4] 2 The Program's economic feasibility analysis compared its estimated economic benefits and costs. The cost-benefit analysis estimated the economic feasibility of the Program by calculating the net present value (NPV) of cost and benefit streams and determining the Program ERR. This includes converting financial cash flows associated with activities contributing to results under RA 2 and RA 3 into economic cash flows to eliminate distortions caused by taxes, subsidies, and other

[4] 1 Ministry of Economy and Finance Report on budget execution and three-year macroeconomic framework published in connection with the 2 finance law.

Cost-benefit analysis monetizes all major benefits and all costs associated with a project so that they can be directly compared with each other as well as to reasonable alternatives to the proposed project.

Benefits include (a) increased water revenues resulting from physical and commercial loss reductions; (b) decreased operational costs resulting from leak detection and reduced energy requirements; (c) revenue from the sale of treated wastewater; (d) GHG emission reductions owing to reduced energy requirements, and (e) reductions in local pollutants from WWTP effluents. Also, the economic analysis considered the incremental O&M savings from reducing production, pumping, and energy consumption costs.

Significant GHG emission savings are expected from water loss reduction programs leading to an overall decrease in energy usage and a reduction in localized pollutants resulting from the increased tertiary treatment capacity of WWTPs. A reduction in localized pollutants is expected from WWTPs upgrades to increase the capacity for tertiary treatment and reuse of treated effluent for green spaces and industrial and agricultural uses. These improvements in effluent quality will decrease pressure on scarce freshwater resources in the receiving water body.

Investments in water loss reduction will postpone the need to invest in non-conventional water sources. Treated wastewater and desalination represent consistent water sources for urban demand and irrigated agriculture, especially in droughts, a diversifying the water supply will increase resilience and water secur Supporting the enabling environment and regulatory framework will provide Morocco with a viable path to securing its water future. Similarly, the principles for water valuation (included under DLI#1) are key to informing water pricing and allocation decisions and promoting more efficient water allocation. Demand management and conservation strategies will be increasingly important as wat scarcity increases to strengthen climate resilience.

The Program is expected to improve the water sector's financial sustainability. In addition to the interventions of the Program, which are expected to increase revenues and reduce operational costs, a financial model of the water sector will be developed to improve financial sustainability and inform policy decisions. It will allow multiple institutions (ABHs, Régies, and water supply operators, among others) to improve the overall financial efficiency of the water sector and public expenditure management by combining capital expenditure planning with water pricing and ongoing operations. Integrating the valuation of water services will allow service providers to consider water pricing strategies across users that enable smart water usage and inform water allocation decisions.

The Program results are intended to support these efforts by fostering accountability, transparency, and trust in the water sector by (a) developing information systems at the national and river basin levels to provide access to regularly updated data to citizens; (b) engaging citizens through communication campaigns to influence perceptions and awareness of water use and improve water conservation; and (c) developing groundwater management contracts with the participation of water users.

Guidance note on shadow price of carbon in economic analysis.

An Integrated Fiduciary Systems Assessment (IFSA) of the Program was carried out following the World Bank Policy on PforR Financing.

The IFSA concluded that the existing fiduciary systems of implementing entities meet Bank PforR Financing requirements, subject to the mitigation measures and actions included in the PAP.

Public financial management -including country procurement systems followed by the entities and assessed by the IFSA- is acceptable to the Bank and meets the requirements for implementing a PforR.

The key fiduciary risks identified in the IFSA include procurement and financial management (FM) risks.

On the FM front, risks include (a) delayed execution of budgets by some ABHs and Régies and commitments of funds due to the challenges derived from the implementation of the PPD and the compliance with E&S requirements; (b) delayed preparation of consolidated financial statements and submission of Program audit reports due to a large number of implementing entities located around the country which requires an effective and adequate fiduciary coordination mechanism; (c) the lack of familiarity with PforR financing instruments of most of the entities (e.g., ABH, Régies, and DRPE) involved in the Program implementation combined with the use of ·Excel· spreadsheets to prepare consolidated financial reports may impact the fiduciary performance of the Program, including the quality and timeliness; and, (d) follow up of audit recommendations and the internal audit unit at the MEE as per the Decree No.05-2019. Bank operations in the sector linked to water and irrigation, such as the Morocco Rural Water Supply Project and the Morocco Large Scale Irrigation Modernization Project, have faced challenges, with an extension of the closing date required to complete works and last payments.

To strengthen the fiduciary coordination, expenditure reconciliation, and financial reporting, the PAP will support (a) the development of standard tools and documents for fiduciary reporting to relevant implementing entities (to be included as part of the POM); and (b) the deployment of performance management with the support of the Bank fiduciary team through monitoring of key fiduciary performance indicators.

Implementing entities will monitor and abide by the World Bank·s debarred/suspended firms list.

An Environmental and Social System Assessment (ESSA) was carried out following the World Bank Policy on PforR Financing. The Program encompasses eligible non-structural and structural activities, with the ESSA identifying measures to strengthen the E&S management system.

Through water reduction programs and WWTPs upgrades to the tertiary treatment level, the Program will support Morocco·s climate adaptation priorities, safeguarding water for domestic, industrial, and agricultural purposes.

The Program will also (a) advocate a participatory and collaborative approach to the implementation of activities using collaborative leadership approaches, which will give boost citizen engagement; (b) introduce positive behavioral changes among stakeholders in terms of management and prudent use of water resources; and (c) contribute to strengthening the grievances redress mechanisms (GRMs).

Potential E&S risks have been identified and can be managed or mitigated.

Following the World Bank·s Operational Policy on PforR, these have been excluded from the Program..[4] 7 The ESSA estimated that the potential negative impacts of additional 22 activities would be reversible, specific risk mitigation measures have been identified, and their implementation will prevent negative impacts.

For those activities that do not require an environmental impact assessment (EIA) according to the EAI Law 12-03 (2003) -such as the rehabilitation of drinking water networks- an ESMP must be prepared before the start of works to ensure that (a) its requirements are included in the works contract; and, (b) that monitoring of the required mitigation measures is effective, documented and evaluated. For activities listed in the EAI Law -such as WWTPs- the EAI Law stipulates that any change to the approved design requires an updated EIA.

The E&S management system is considered adequate within the context of the PforR Policy, and some actions will be included to strengthen its performance.

.), identify adequate social and environmental mitigation measures, and prepare and implement the corresponding action plans based on the following:

- Environmental and Social Impact Assessments (ESIAs), including resettlement action plans as necessary (RAPs), for all activities involving moderate E&S risks
- Environmental and Social Management Plans (ESMPs) for all activities involving moderate E&S risks
- Screening checklists for all activities with low E&S risks
- Community participation plans and stakeholders· engagement plans as required by the ESSA Action Plan and as per the E&S Technical Manual
- Establishment of functional and accessible GRMs for all project-related activities as stated in the ESSA Action Plan.

The capacity for managing E&S aspects related to the Program activities requires significant strengthening measures. The EIA Law 12-03 provides the administrative authorizations for any project subject to a decision of environmental acceptability based on an EIA. The procedures and principles governing the context of the EIAs are generally in line with international practice, and the environmental management and EIA procedures are clear at the technical level and sound at the institutional level. However, the capacity of the various institutions responsible for managing E&S aspects related to the Program activities requires significant strengthening. Thus, the Program will support specific measures to strengthen the performance of the Moroccan E&S management systems, including aspects related to (a) E&S management systems; (b) M&E; and (c) E&S management capacities

The overall risk rating for the Program is Substantial, due to substantial risks associated with the macroeconomy, sector strategies and policies, institutional capacity for Implementation and sustainability, fiduciary, E&S, and stakeholders (see datasheet).

The New Development Model and the reform agenda launched by the GoM would improve the attractiveness of Morocco for private investment, especially in promising sectors with high value-added, and mitigate the impact of these potential risks. The IMF has recently approved a precautionary Flexible Credit ine, which will boost Morocco·s external buffers contributing to mitigate residual macroeconomic risks.

Reducing the gap between water demand and supply, while protecting and securing water resources and building resilience to increasing hydro-climatic variability due to climate change, is a priority of the GoM. Activities contributing to the Program objectives of mobilizing water resources while strengthening the capacity for improved water resources planning and management are also key to climate change adaptation as per the country·s NDCs. Mitigating risks related to data sharing and access are central to the Program, including developing standardized data-sharing protocols and facilitating data transfer, as a share of the Program·s success depends on data-sharing and data-access policies of the ministries involved and ABHs. The creation of SRMs after the passing of the Law in Parliament (expected in 2023) may impact Program implementation if it involves Régies included under the Program.

Since the regionalization process is expected in the coming years, specific mitigation measures will be identified during Program implementation as needed.

The Program also supports a series of important institutional reforms, including measures to improve the valuation of water and financial sustainability.

The investment capacity building under the Program will establish a strong institutional basis for the sustainable continuation of the systems and services.

The increasing investment in non-conventional water resources, particularly desalination, is also changing the sector's cost structure.

A series of measures have been integrated into the results framework to support the process of building consensus on the roadmap required to ensure the institutions and the financial structure of the sector are sustainable, including through the adoption of the PNE.

These reforms will be supported by aligning with the high-level inter-ministerial mechanisms established under the PNAEPI.

The Program is expected to bring significant E&S benefits by strengthening sector institutions and increasing water availability.

Construction activities required to achieve the Program's objectives are largely limited to small-scale, localized works related to reducing water losses in distribution networks and upgrading existing WWTPs and associated infrastructure for reuse.

This introduces several risks to achieving the Program·s intended outcomes but reflects the complexity of the various measures needed to improve water security along the water value chain. This includes stakeholders involved in managing the development of water resources, those involved in the provision of services, along with a wide range of users in various sectors. These challenges define the objectives and inform the design of the activities supported under the Program and the expected outcomes, part of which is focused on strengthening institutions in the water sector. Provisions have been included in the implementation arrangements to strengthen this coordination, including budgeted technical assistance for the PCU and focal points from the different implementing entities. Specific results are included to incentivize collaboration, including technical working groups for the institutional and financial sustainability aspects · which will be adopting innovative approaches such as collaborative leadership.

This indicator monitors the preparation and adoption of the National Water Plan (PNE).

The MEE prepared a draft PNE submitted for review by the Interministerial Commission for Water in December 2019.

It shall reflect the increasing uncertainty derived from climate change; and the principles to strengthen the sector·s governance, institutional, and financial aspects. The revised draft PNE shall be prepared using participative approaches, as provided by the Water Law, including contributions of the MEF, MEE, MoI, MA, OCP, service providers (irrigation and water supply and sanitation services), and others.

This indicator is achieved as follows:

Year 2025: Two (02) ABHs included in the Program have developed or upgraded their water information system for measurement and management of water quantity and quality.

The volume of water savings are calculated in cubic metres as per the following formula:

[[Efficiency of the network in year N+1] · [Efficiency of the network in year N]] X [volume of water sold in year N+1].

for reuse purposes, the distribution network, and associated infrastructure (pumping stations, storage tanks, electric line, and others).

2028: For the three (3) aquifers included in the Program, three (3) participatory groundwater management contracts are signed according to the regulations in force.

This result is scalable up to 10 percent of 50 of smart meters for monitoring groundwater withdrawals by large users installed in Program areas during the Program implementation.

The annual action plan shall include - at a minimum- defined objectives and priority areas, costed actions, identified sources of financing, monitoring indicators, and an implementation timeline

The annual action plan shall include - at a minimum- defined objectives and priority areas, costed actions, identified sources of financing, monitoring indicators and an implementation timeline.

Year 2025: The draft Decree for establishing the SNIEAU and ABH water resources information systems is consulted according to the regulations in force.

Year 2026: The Decree for establishing the SNIEAU and ABH water resources information systems is adopted and published in the Official Gazette according to the regulations in force.

Year 2024: A diagnostic of the hydrological and hydrogeological data measurement chain is prepared by the MEE.

The data measurement chain encompasses:
(a) Data acquisition (measurement, gauging, data collection)
(b) Data transmission and storage (processing, treatment, analysis, control, validation, and storage)
(c) All other related data organizational and regulatory aspects

Year 2025: The action plan and guidelines/manuals to improve hydrogeological and hydrogeological measurements are prepared by MEE.

AND

The water data management system is developed. The water data management system shall contain, at a minimum, the necessary functionalities and modules for data acquisition, processing, and storage based on the diagnostic and action plan produced in prior years.

Year 2026: A quality and control assurance plan for hydrological data is prepared by the MEE.

The regulatory texts shall be in accordance with the manuals/guidelines prepared in year 2025 related to article 129 of the Water Law 36-15, specifically to the ·carrying out of measurements, observations, inquiries and investigations·, and include a Decree and number of orders/general circular concerted with the relevant agencies and procedures as defined in the POM, and issued to the relevant agency for approval.

Year 2027: The quality and control assurance plan for hydrological data is adopted by the MEE. Also, workshops for capacity building on all aspects of improving the quality of hydrological and hydrogeological data measurement are carried-on in at least one (01) ABH under the Program.

In addition, hydrological or hydrogeological data produced by an ABH has been certified, hydrological and hydrogeological data are measured, collected, transmitted, and stored in accordance with guidelines/manuals and procedures prepared for quality assurance and quality control.

A ·new station· is defined as a station constructed and equipped under the Program, and includes hydrological, climatological, or pluviometric stations.

The studies for the stabilization of gauged sections shall include, at a minimum, the minor civil works design and topographic evaluation or technical specifications for the stream cross sections of the gauged stations to be stabilized.

Year 2026: 12 new or existing stations in the ABHs included in the Program are producing and communicating relevant data for water resources evaluation to the ABH and MEE.

Year 2027: 12 new or existing stations in the ABHs included in the Program are producing and communicating relevant data for water resources evaluation to the ABH and MEE.

AND

30 new or existing piezometers in the ABHs included in the Program are producing and communicating relevant data for water resources evaluation to the ABH and MEE.

This indicator is achieved as follows:

2024: The MoI has adopted by Circular the minimum service standards for electricity and water supply distribution and wastewater collection and treatment to be complied with by multiservice operators in charge of electricity and water supply distribution and wastewater collection and treatment.

2028: An annual report produced by the MoI on multiservice operators' performance, including comparative KPI benchmarking, is available in the SNIEAU.

Year 2025: The financial model for the water sector is endorsed by the Technical Working Group, under the MEF leadership.

The financial model shall encompass water mobilization, water resource management at national and local levels, service delivery (potable water production and distribution, wastewater collection and treatment, and irrigation services), wastewater reuse, and flood control. The financial model shall capture CAPEX and OPEX and the different revenues, including national transfers and local contributions, taxes, and users' contributions (tariffs and fees)-.

Year 2026: Based on the financial model developed in 2025, the Technical Working Group endorses a draft financial sustainability action plan for the water sector.

Year 2028: The financial sustainability action plan for the water sector is adopted by the Interministerial Commission of Water.

defined as a hydraulically isolated area within the larger water distribution network where all inflows and outflows are monitored, and therefore a reliable water balance is known.

Year 2025: The draft Decree on seawater desalination is consulted with relevant stakeholders according to the regulations in force

Year 2026: The Decree on seawater desalination is adopted according to the regulations in force.

AND

The draft bylaw defining the norms for wastewater reuse for irrigation purposes is consulted with relevant stakeholders according to the regulations in force.

Year 2026: The draft decree on the modalities for wastewater reuse is adopted according to the regulations in force.

This indicator monitors progress in the signature of agreements for reusing treated wastewater within the Program Area.

2026	4,000,000.00	8,000,000.00	500,000 per every additional 250,000 m3 potable water saved through the implementation of NRW plans by Regies in Program area.
2027	3,250,000.00	6,500,000.00	500,000 per every additional 250,000 m3 potable water saved through the implementation of NRW plans by Regies in Program area.
2028	3,500,000.00	7,000,000.00	500,000 per every additional 250,000 m3 potable water saved through the implementation of NRW plans by Regies in Program area.

2026	3,400,000	3,400,000.00	500,000 per every additional 500,000 m3 of treated wastewater made available for reuse under the Program Area.
2027	11,900,000	11,900,000.00	500,000 per every additional 500,000 m3 of treated wastewater made available for reuse under the Program Area.

The MEE prepares consolidated semi-annual report based on the information provided by the ABHs under the Program area indicating the number of new smart meters for monitoring groundwater withdrawals by large users that have been installed in the Program area over the previous six-month period and submits it to the PCU with the supporting evidence.

For the DLR#4.2 - the verification procedure is as follows:
The MEE sent to the PCU evidence of Water Information Systems operationalized with accompanying documentation and submits it to the IVA.

The IVA review that SNIEAU complies with the minimum criteria established in the definition of the indicator and in the POM (as relevant)
For DLR#4.4 the MEE sends to the PCU evidence that two (2) ABHs have published the annual hydrological status, including water uses and the evolution of the water quality within the jurisdiction of the ABH in the SNIEAU and submits it to the IVA to review that the minimum criteria established in the definition of the indicator and in the POM (as relevant).

The IVA analyses the semi-annual consolidated reports to verify that:

(a) In 2024, the MoI has adopted by Circular the minimum service standards for electricity and water supply distribution and the wastewater collection and treatment to be complied with by multiservice operators in charge of electricity and water supply distribution and the wastewater collection and treatment.

(e) An annual report produced by the MoI on multiservice operators' performance, including comparative KPI benchmarking, is available in the SNIEAU.

The semi-annual progress report shall be accompanied by the supporting annexes that demonstrate the participatory approaches, and the applied methodology.

(b) In 2025 · the financial model for the water sector is endorsed the Technical Working Group.
(c) In 2026 · the financial sustainability action plan for the water sector is endorsed by the Technical Working Group.
(d) In 2028 · the financial sustainability action plan for the water sector is adopted by the Interministerial Commission for Water or any other consultative body.

The MoI prepares annual consolidated progress reports on the water savings (disaggregated by Regies) with accompanying documentation on the water savings achieved by each of the regies included in the Program and submits it to the PCU and the IVA accompanied by the audited annual management reports (rapports de gestion) of the Regies included in the Program.

Each year, the IVA analyses the MoI consolidated reports and the audited annual management reports to verify the calculations of the water savings indicated by the MoI in the consolidated report.

The IVA conducts field verification on a representative sample of sites to determine consistency between progress and actions reported in the annual management reports and physical progress in the field.

For the revised bylaw defining the norms for wastewater reuse for irrigation purposes adopted by the MEE, the progress reports are prepared by the MEE.

The MoI ensures that the Convention Monitoring Committee for each reuse project is established in accordance with the provisions of the treated wastewater reuse agreement.

The construction companies selected for the implementation of works included in the treated wastewater reuse agreement prepare semi-annual progress reports that are submitted to the Convention Monitoring Committee and include · at a minimum- (a) physical and financial progress of the construction of the reuse project, (b) records of environmental and social management, (c) grievance management, and (d) incident/accident reports.

Once the Reuse Project has been completed, the convention Monitoring Committee prepares a Consolidated Final Project Report, which includes · at a minimum- (a) evidence of the physical completion of works included in the treated wastewater reuse agreement; (b) final financial balance; (c) evidence of compliance with the quality standards for wastewater to be reused for the specific use established in the treated wastewater reuse agreement; (d) summary of environmental and social management, (e) summary of grievance management, and (e) summary of incident/accident reports.

The MoI-DRPL submit semi-annual reports to the IVA and the PCU indicating for completed projects - the volume of treated wastewater effluent made available for reuse during the six months, accompanied by the Consolidated Final Project Report prepared by the Convention Monitoring Committee, and for projects under implementation · summary of the progress made during the six months under the treated wastewater reuse agreement signed and not yet completed.

It will also conduct random treated wastewater quality sampling to verify they comply with the relevant Moroccan regulations on water quality for reuse.

For DLR#8.2: The revised bylaw (arrêté) defining the norms for wastewater reuse for irrigation purposes has been adopted by the MEE - the MEE sends to the IVA evidence of the adoption of the bylaw according to the regulations in force.

This was due primarily to non-climatic factors, such as population growth in the north, the expansion of irrigation, and urban, industrial, and tourism development, with water demand already exceeding available supply. Based on the currently available sources of supply in the water accounts (i.e., freshwater, treated wastewater reuse, and desalinization), the annual average inflow is estimated at around 18 billion cubic meters (BCM) per year, with around 14.5 BCM per year available and sustainably exploitable with existing infrastructure, including 4 BCM per year of renewable groundwater resources.

The current water deficit is estimated at 1.8 BCM annually, with demand expected to increase by 15 percent between 2020 and 2050.[49] Shortages are aggravated by the loss of existing storage due to high sedimentation rates in the country's portfolio of large dams, estimated at nearly 75 million cubic meters (MCM) per year, and water pollution in many areas.

These challenges are compounded by the unequal distribution of water over space and time, with 70 percent of the surface water resources concentrated in the country's northwest, representing a mere 15 percent of the territory.[50] Surface water shortages have led to the over-exploitation of groundwater, with current groundwater withdrawals exceeding the exploitable level by 28 percent (or 1.1 BCM per year), increasing energy-related greenhouse gas (GHG) emissions due to the need for deeper groundwater pumping.

Between 1960 and 2020, the Kingdom built more than 120 large dams, a tenfold increase in total water storage capacity, bringing the total to 135 large dams with a total capacity of 17.5 BCM. A further 100 small dams have been constructed to provide another 100 MCM to meet local water, irrigation, and livestock needs. There are also 13 water transfer systems between river basins to increase resilience and assurance of supply. Since 1995, the GoM has made significant investments to address the gap through the development of desalination plants, with the installed capacity reaching 220 MCM per year by the end of 2022. While desalinization represents less than 2 percent of all mobilized water resources, it provides a high level of assurance, with 44 percent for potable water, 41 percent for irrigation, and 14 percent for phosphate production.

Increasing water management efficiency is critical to close the water demand-supply gap[51] Regarding potable water , reducing existing water losses in the transport and distribution networks could lead to potential water savings of 400

[49] Total water demand by 2050 is expected to be between 18.7 and 20 BCM per year taking climate change into account. Water demand from urban households is expected to increase by 50 percent (from 1.1 to 1.7 BCM per year) during the same period, with corresponding increases in effluent quantities.

These water management investments will reduce demand by about 2.2 BCM per year.

[57] A permanent reduction in water supplies could significantly affect the Moroccan economy,[58] with water scarcity exacerbating competition among water users and leading to a 6.5 percent decline in real GDP under extreme conditions.[59] For example, hotter and drier conditions are expected to increase the water requirement of crops by up to 12 percent, increasing demand for irrigation and further stressing limited water resources. While infrastructure will be required to help adapt to the changing conditions, the potential magnitude and uncertainty associated with future climate conditions will also require ambitious policy reforms that reflect the value of water resources, increase the transparency of costs along the water value chain, incentivize more efficient and rational uses, and balance competing demands.

The GoM is implementing comprehensive large-scale water sector investment programs to address the country's water challenges.

The PNAEPI promotes supply-side solutions, with a significant portion of investments aimed at increasing the mobilization of water resources by about 3 BCM per year by 2027, with a strong emphasis on non-conventional water resources. The PNAEPI establishes targets for reducing water-losses, acknowledging the incremental cost of new water supplies, and includes provisions for the reduction of dam sedimentation, incentives for

[52] Ministry of Equipment and Water, 2019.

Water-loss targets in transport and distribution networks should also reflect the interdependency of upstream and downstream users, and the valuation of water and its externalities.

[54] The implementation of the Irrigation Water Conservation Program 2008-2020 resulted in the conversion of 462,000 hectares to localized irrigation (395,000 hectares for individual reconversion and 67,000 hectares for reconversion of collective land (including ongoing works), intensifying agricultural production through more reliable irrigation and by doubling water productivity on average (measure as US$/m^3 For instance, in the Doukkala ORMVA, the program contributed to increasing water productivity, from 5.7 to 10.6 MAD/m^3 in 2018; and, to increase the share of high-value-added crops (vegetables, fruit orchards, and oilseeds) relative to low-value-added crops (cereals, beets, and fodder).

The legal framework for the water sector in Morocco has evolved in response to the changing socio-economic, political, and resource realities.

[60] Law 10-95 (1995) stipulated the rules for integrated, decentralized, and participatory water resources management to guarantee the right of citizens to water. The foundations introduced by the Water Law 10-95, aiming to improve water use efficiency, provide universal access, reduce disparities between rural and urban areas, and ensure water security across the country, were kept and reinforced by the Water Law 36-15 (published in 2016). Also, the Water Law 36-15 further emphasized decentralized, integrated, and participatory management and planning of water resources; strengthened consultation and coordination bodies and organizations by establishing water basin councils; established the legal foundations to diversify sources of supply through the use of unconventional water resources; introduced water-related information systems; and, strengthened the institutional framework and mechanisms for the protection and conservation of water resources- including participatory groundwater management contracts. [61]

[60] Doukkali, M.R.

The institutional framework for water resources management includes various national, regional, and municipal actors.

Under Royal instructions, a commission (Commission Interministerial sur l·Eau,[3] was created in 2017 to formulate an emergency plan to confront the drought that started in 2015 and accelerate the mobilization of additional water resources for potable water and irrigation purposes. Water resources planning and mobilization at the national level is the responsibility of the Ministry of Equipment and Water (MEE)[4], in coordination with all relevant sector stakeholders. River Basin Agencies (Agence du Bassin Hydraulique, ABHs), created by the Water Law 10-95 as technical and financially independent entities, are responsibility for planning and managing water resources at the basin level based on the principles of integrated water resources management and monitoring water use and quality.[5] The River Basin Councils, introduced by the Water Law 36-15 and established in 2022, are in charge of revising the PDAIREs, represent the interests of the regional and municipal authorities and water users (services providers, farmers, and public and private associations in the water and climate domain). Regional Agricultural Development Offices (ORMVAs) provide irrigation services withing a defined irrigated perimeter (usually encompassing large-irrigation schemes), with irrigation covering roughly 1.7 million hectares (accounting for 19.4 percent of the cultivated area), representing about 45 percent of the agriculture value, and contributing 75 percent of the total agriculture exports of the country.

The institutional framework for delivering potable water services encompasses actors at three levels of government. The National Office for Electricity and Potable Water (Office National de l·Electricité et de l·Eau Potable: ONEE) is a public enterprise responsible for potable water planning and has traditionally been the major national potable water producer, with their tariff being regulated by the Interministerial Commission of Prices (headed by the MEF). In connection with the GoM·s strategic scaling-up of non-conventional water resources, the OCP Group has been tasked by the GoM with developing desalination plants for phosphate production and providing potable water in neighboring municipalities. The MoI, through its Direction of Public Local Networks (MoI-DRPL), plays a supporting role in planning, implementing, and monitoring the distribution of potable water and wastewater collection, which are the responsibility of the municipalities, exercise control over private concessions, and set retail tariffs for water and wastewater services. Hence, they may provide the service directly or by delegation to autonomous municipal public utilities (Régies)[7], the ONEE, or private concessionaires. The ongoing reform of the water supply sector will entail the introduction of regional multiservice companies (Sociétés Régionales Multiservices, SRM) for the provision at the regional level of energy and water distribution services and wastewater collection and treatment (see more below under RA1).

[4] MEE is responsible for initiating, promoting, and coordinating the protection of water resources, pollution abatement and legislation enforcement. In addition, it is responsible for environmental control, auditing, and reporting, including public awareness and participation regarding water resources and the environment.

[5] Their duties include preparing PDAIREs, authorizing water abstractions and discharges, and maintaining a public register; collecting fees for abstraction and effluent discharges; providing financing and technical assistance for water pollution prevention; ensuring efficient water use; monitoring water quality; enforcing laws related to water resource protection; setting up an emergency response system; and promoting public awareness about water resource management.

[6] Evaluating the contribution of irrigation to economic and agricultural growth: the case of Morocco.

[7] The Régies are legal municipal public entities, with financial autonomy, under the supervision of the MoI. There are currently 12 autonomous distribution boards: seven for water, electricity, and wastewater distribution; and five for water and sanitation services distribution · of which 11 are included under the Program.

Through tariffs and fees, user contributions are generally insufficient to cover operational and maintenance costs. Bulk and retail tariffs are not based on production and distribution costs. The under-collection of withdrawal and water pollution fees further undermines the sector·s financial sustainability. [6] 8 Between 2012 and 2017, the water basin organization collected less than US$7 million per year on withdrawal and discharged fees, of which discharge fees represented less than 4%. Low water tariffs could present barriers to realizing the GoM·s efforts to boost water security, including further development of water supply sources (dams and water transfers) and the scale-up of non-conventional water resources (desalination and wastewater reuse).

The changing nature of the water balance and climate change challenges require institutional realignment to boost sector resilience to future uncertainty. The sector has changed its focus to respond to water challenges from emphasizing public works and improving infrastructure to sustainable resource management and increasing water security.

As the sector transitions in response to increasing water scarcity and future climate uncertainty, the institutional realignment of roles and responsibilities is needed to ensure affordable and sustainable water security solutions.

These include support for the GoM·s efforts to strengthen the governance of the water sector to ensure that it is fit for purpose and can deliver on Morocco·s expectations related to the

[6] 8 Source: Cour des Comptes

The activities related to groundwater management and participative aquifer management contracts will be only implemented within the geographical perimeter of action of the six ABHs included in the Program.

The activities related to the signature and implementation of conventions for the use of treated wastewater will b implemented within the geographical perimeter of action of the six ABHs included in the Progr

Strengthening Water Sector Governance

13. The technical assessment highlights the need to re-align institutions to ensure good water governance in response to changes in the water matrix, mobilization costs, and the increasing role of the local and regional governments.

Also, given the ongoing reform on the distribution of water services, the Program provides an opportunity to support the improvement in performance benchmarking of public-owned service providers for electricity and water supply distribution and wastewater collection and treatment. At the decentralized level, the technical assessment found that adopting performance targets and fit-for-purpose built-in capacity can improve ABHs' performance and greater effectiveness in managing, protecting, and planning water resources within their respective domain, including financial autonomy.

The impact of drought and floods and the pressure of growing demand are all factors that require planning for water resources development and management strategies. Thus, since the 1970s, the GoM has undertaken efforts for water resources planning aimed at meeting Morocco·s water needs, with the objectives of (a) integrated planning and conjunctive management of surface and groundwater; (b) optimizing the allocation of water resources to meet present and future demand in the medium and long term in support of Morocco·s economic and social development growth at national and local levels and also between sectors; (c) providing access to water for the various regions of the country to ensure balanced development and the promotion of water-poor regions through water transfers from surplus to deficit regions; and, (d) protecting water resources.

Water Planning Process.

According to the Water Law 10-95, the PNE sets the priorities for the mobilization and use of water resources based on the PDAIREs and defines the conditions for water transfers between river basins. Based on Water Law 36-15 requirements, six PDAIREs have been approved by the Board of Directors of the ABHs, and four more are expected to be approved by December 2024. The PDAIREs include (a) a quantitative and qualitative assessment of water resources; (b) water demand by sector and use category and water balances and water resources development schemes; (c) the definition of the investments at the river basin; (d) the economic and environmental evaluation of the proposed schemes/investments; and, (e) the implementation modalities of these water resources development schemes/investments.

Based on the Water Law 36-15, the MEE prepared a draft PNE and submitted this for review by the Interministerial Water Commission in December 2019. Given the increasing risk of water insecurity and the renewed vision for the country outlined in the New Development Model Report, the existing draft PNE is under review.

The Program will incentivize revising the existing draft, its endorsement by the key consulting bodies (Interministerial Water Commission and Superior Council of Water and Climate), and its adoption by Decree (under DLI #1). The revised PNE, which will set the 30-year vision for the water sector, the draft shall encompass the following:

- A comprehensive diagnosis of the water sector
- An assessment of the major sector challenges and threats -including the impacts of climate change- and potential scenarios of evolution of demand and supply under several climate scenarios
- The long-term vision, objectives, and strategic orientations of the water sector
- The priorities for water resources mobilization and their use and the protection of water resources (surface and groundwater)
- The principles to strengthen the sector's governance, institutional, and financial aspects for water security · including principles for water valuation and cost recovery
- Its monitoring and implementation mechanisms

18. The PNE revision and adoption entails significant coordination efforts among relevant stakeholders, including the MEF, MEE, MoI, MAPMDREF, OCP Group, and service providers (irrigation, water supply, and sanitation services). It reflects the water needs of sector-specific plans (notably the Green Generation and OCP Group Water Plan). The Superior Council of Water and Climate, chaired by the Prime Minister, reviews, and provides its advice on the PNE, which is adopted by Decree of the MEE.

The Organic Law 113-1 (201) specifies that the municipality ·creates and manages the public services and facilities necessary for the offer of local services in the following areas: ·· the distribution of water and electricity; public lighting; · liquid and solid sanitation and wastewater treatment plants· (Article 83). The Municipal Council of the Commune decides the management method of public services delivery under its jurisdiction and by the laws and regulations in force. As such, municipalities are responsible for defining service objectives, monitoring, and controlling the execution of the service, and directing investments. The municipalities shall define master plans for sanitation and wastewater management. Municipalities exercise their functions under the tutelle technique of the MoI, which provides technical assistance to the multiservice operators. They coordinate at the national level the plans for developing municipal public services, including their pricing.

By the provisions of article 143 of the Constitution, the region has an essential role concerning other local authorities in the preparation, implementation, and monitoring of regional development programs and regional spatial planning schemes while respecting the competencies of other local authorities. The region has a legal personality and administrative and financial autonomy to carry out these tasks. The regions can set up Local Development Companies (Societies de development local), which are local public companies whose capital is controlled by the local authorities, which can serve the objective of the regions to implement their responsibilities for equipment and infrastructure with a regional dimension. Regionalization offers an opportunity to revisit the formulation of the national sectoral strategies to anchor them better in a local development process that considers the specific regional advantages and needs.

Greater regionalization necessitates significant reforms in the water sector, given the more prominent role of the regions.

Regions have a significant role in the rollout of significant reform to distribution services. The reform would entail the separation of bulk from distribution services, with bulk energy and water production handled by a national-level operator; and water supply and energy distribution and wastewater services (collection and treatment) entrusted to SRMs.

The GoM is reforming the distribution segment of the water supply and wastewater-collection service delivery chain.

In July 2021, a memorandum of understanding (MoU) was signed between the MoI, the MEF, and Ministry of Energy, Mines, and Environment, and the ONEE to launch the process of creation of regional multiservice companies for the distribution of electricity, drinking water, and wastewater collection and treatment at the level of each region. In December 2022, the draft Law for the creation of SRM was presented to Parliament, and it is expected to be approved during 2023.

Currently, Régies · as autonomous subnational entities- are obliged to submit an annual audit to the MoI, which includes reporting the key information regarding operational and financial plans. The Program will support the establishment of reporting requirements, with greater regularity, to be submitted through a digitalized performance information system to be managed by the MoI.

Also, the Program will support the definition of minimum service standards for electricity and water supply distribution and wastewater collection and treatment.

Groundwater management remains a challenge due to over-abstraction and the lack of data (its quality and quantity).

There is no assessment of the available groundwater reserves, a projection of declining production, or an estimate of the cost increase of pumping and quality deterioration for overused groundwater.

Groundwater has come under unprecedented pressure over the past thirty years due to repeated droughts in the country, which have greatly reduced surface water supplies, pushing irrigators to draw more from aquifers, and partly because of the increase in water needs induced by the country's socio-economic development. The overexploitation of groundwater leads to additional costs for pumping and deepening wells/boreholes, which is problematic for small farmers who practice subsistence agriculture.

The Program provides incentives in three key areas to improve groundwater management. Since 2007, the GoM, through the MEE and the ABHs, has established participatory aquifer management contracts to manage groundwater resources better. These contracts are designed to define the organizational, technical, and financial measures that shall be in place to sustain a given aquifer and avoid over-exploitation by the various aquifer users. Through DLI#2, the Program provides incentives to strengthen the regulations for aquifer management contracts, sign aquifer management contracts for critical aquifers, and monitor groundwater abstraction by deploying smart meters. The Program will support the preparation of the regulatory provisions for participatory aquifer management contracts according to Water Law 36-15. Studies and groundwater knowledge generated under the Programs and supported by strong scientific reviews will also be the basis for preparing and adopting participatory aquifer management contracts for the Berrechid, Maamoura, and Bahira aquifers, which are currently overexploited and for which there is a need to deploy an aquifer management contract.

Since early 2023, the ANCFCC has been implementing an economic cadastre, which will gather

[7] 0 Geosciences 2020, 10, 81, Moroccan Groundwater Resources and Evolution with Global Climate Changes, Mohammed Hssaisoune, Lhoussaine Bouchaou, Abdelfattah Sifeddine, Ilham Bouimetarhan, and Abdelghani Chehbouni.

Thus, the ANCFF economic cadastre will support a greater knowledge of groundwater and would serve as a basis for launching wells-legalization campaigns to register unauthorized wells and control better groundwater abstractions. Under DLI#4, the SNIEAU will be interoperable with the ANCFCC economic cadastre platform. Hence, data will be available to water entities (at national and local levels) to better information decisions on groundwater management.

Finally, the GoM is deploying smart meters for large water users to better manage groundwater withdrawals. The Program will support these efforts through DLI#3 by disbursement against the number of smart meters for large water users installed in aquifers within the Program area · that is, for the three aquifers for which aquifer management contracts will be signed as part of the Program and others.

The ABHs were created by Water Law 10-95 to ensure integrated water resources management at the basin level. They are independent entities with financial autonomy under the tutelle technique of MEE. The Water Law 36-15 adoption in 2016 solidified this decentralized approach by creating water basin councils and strengthening the main obligations and roles of ABHs. As per Water Law 36-15, the mandates of the ABHs include monitoring, protecting, managing, and planning water resources, developing, and supplying water, monitoring and controlling water use and quality, and flood protection within their respective basins.

The main challenges across ABHs pertain to insufficient information management systems informing the preparation and implementation of the PDAIREs, operation of water infrastructure, water allocations, and flood forecasting. Insufficient revenues and human resources also prevent ABHs from monitoring and controlling water abstraction and pollution efficiently, undertaking watershed and flood protection, ensuring dam O&M and safety, and sustaining reliable water sources. This is also due to the need to implement more cohesive actions for the control of the Hydraulic Public Domain and of specific regulations in place on the delimitation of the Hydraulic Public Domain, water police, and permissible effluent discharge values.

Existing frameworks mainly focus on financial performance and remain limited to support prioritization efforts for overall ABH strengthening.

A solid base of water sector information is required to support efficient and resilient sector development. The technical assessment identified actions to improve water sector data and information, which need a holistic approach that encompasses all systems that make up for water information management systems to facilitate access and bring more utility of information services to decision-makers and users. These systems include monitoring and observation networks, data management systems, and information services delivery and feedback systems. Improving water information services in Morocco requires increasing the spatial coverage and density of observation networks. These systems require quality management systems to ensure water data reliability for different applications. Developing information systems to boost the exchange, public access, and utility of water information to users through specific products and services will support efficient water resource management and planning.

Water observation and monitoring networks.

The ABHs manage hydrological and hydrogeological networks, while the strategic development of these systems at the national level lies under the MEE-DRPE. The manual and automated data measurement and transmission from existing networks further face shortcomings due to deficiencies in data quality assurance practices and supporting Information and Communications Technology systems. Thoughtful operation and maintenance and sufficient financial and human resources are necessary to ensure these systems' sustainability over time.

The Program will support the development of hydrological and hydrogeological networks in the ABHs included in the Program. Establishing new hydrological stations and piezometers will extend existing measurement networks to areas not sufficiently covered within these basins. The maintenance of existing stations, including the modernization of select stations and piezometers with telemetry equipment, will further support the rehabilitation and upgrade of these systems. The development of these networks will be implemented following a diagnostic and action plan that will enable the adoption of a more strategic approach based on assessed water data needs, benchmark best practices, and cost-effective solutions. Implementing these plans will also include the training and capacity development of ABHs and MEE technical staff to operate and maintain improved systems, including optimizing their management and O&M.

Data management systems and water data quality assurance. The existing water resources database (BADRE21), which encompasses different data processing and storage modules, does not meet the functional needs of ABHs and MEE. This database lacks automatic synchronization between ABHs and central-level data management systems. The backbone software making up this database further requires upgrading to reflect functionalities with current practices for data visualization, consolidation, and protection. In addition, standard procedures for undertaking surface and groundwater measurements, including specific tools and technical capacity for data quality assurance and control across the data measurement chain,[72] remain underdeveloped. These are necessary to maintain a satisfactory level of quality in a data set or data consolidation to build institutional data trust and make water data and information sufficiently reliable to potential users.

The Program will support the development of an improved data management framework to operate for the processing, storing, and consolidating of water data at the ABH and central level (DPRE-MEE). This system will include improved data storage modules (databases) specific for surface water, groundwater, and flood data, each with Geographic Information Systems (GIS) capability.

The Program will also support the preparation and adoption of guidelines and manuals detailing the procedures for surface and groundwater measurements and the operationalization of tools and practices for data quality assurance and control by ABHs included in the Program. Adopting quality management principles will enable ABHs to enter a formal certification process. Quality assurance and control processes will cover the life cycle of the water data, from the equipment installation and observation through its storage and use.

In Morocco, water information systems at the ABHs level constitute the basis for data consolidation at the basin level. The Water Law 36-15 mandated the creation of a National Water Information System (SNIEAU) as the chapeau framework to consolidate basin-level water data at the national level.

This is partly due to specifications of the Water Law 36-15 related to the SNIEAU and the functions and procedures for water data (Articles 129 and 130) requiring prior definition and formalization. Water information systems also require further development of decision-support functions for water resources planning and management, including specific data products or services to decision-makers and users.

Improved water information management systems will enable the reliable access and use of water data and information for decision-making at the basin and national levels. To this end, the Program will support the adoption of the decree defining all specific requirements for establishing the SNIEAU · following the Water Law 36-15 specifications.

The Program will also incentivize the operationalization of water resources information systems at the basin and national levels, involving their necessary upgrades to meet compliance with the established norms, the interoperability among systems for the consolidation of water data from basin to national level systems and with other sector systems, and to enable and ensure publicly and reliable access of water data and information to stakeholders and users. The access and use of water data and information will be further enhanced by developing and upgrading specific decision-support tools for water quality, groundwater monitoring, and the generation of river basin water accounts.

Improving Financial Sustainability and Water Use Efficiency

39.

Tariffs for water supply and irrigation services are not

[72] Include the ensemble of processes and means for data measurement and collection, treatment, transmission, processing, and storage.

These challenges and changes in the sector's cost structure call for a comprehensive approach to the valuation of water that can inform future-focused policies and regulations to improve financial sustainability. These policies and regulations will be important to reveal the range of values among different stakeholders and develop a shared vision within communities, to optimize the sustainable and efficient allocation of water, ensure the equitable pricing of water services, and guide investments aimed at safeguarding water security, its contribution to economic growth and development, and maintaining social equilibrium.

The Program will support the Government·s efforts to anticipate and address the challenges of financial sustainability in the sector. The NDM report recommends measures ·reflecting the true value of the water resource and incentivizing a more efficient and rational use and management of the resource.· The goal is ·future-orientated· policy processes that anticipate, rather than react to, new challenges and account for the change in socio-economic and environmental conditions, shifts in societal values, and pervasive uncertainty associated with climate change. As part of updating the PNE, the Program will support a qualitative process that can build consensus around national priorities and reveal local values, culminating in adopting principles that will inform the development of quantitative tools to improve financial sustainability. The Program will build on these principles to identify and update the underlying costs and benefits associated with investments in the water sector to inform the development of a consolidated financial model.

This information is critical to informing and guiding the process of developing pricing strategies for the various sub-sectors that can be used to create incentives for better water use and consumption while safeguarding water for the environment and other intangible value allocation options.

A key pillar of the PNAEPI is the reduction of water losses and service providers· performance improvement to enhance Morocco·s water security. In 2022, water losses (physical and commercial) varied from 19.9 to 31.28 percent among the Régies included in the Program. Given Morocco·s water scarcity challenges and the increasing cost of water mobilization, a further increase in potable water distribution efficiency is critical to enhance water security and support the financial sustainability of service providers.

The Program will support the implementation of the water loss reduction plans prepared by the selected Régies in the Program area.

In addition, existing DMAs will be upgraded, where necessary, with electromagnetic flow meters and remote management capabilities, superimposition of commercial and hydraulic sectors, installation of leak detection equipment (e.g., acoustic sensors), replacement of old and defective meters, pressure modulation, geo-localization of water connections and migration into GIS platforms. These actions will help improve the monitoring and calculation of water balances in each DMA, increasing the efficiency of leak detection and decreasing water losses In addition to these actions, the Program will support repairing, rehabilitating, and replacing old pipes, depending on their damage status.

The water loss reduction activities supported under the Program could lead to water savings of 20 MCM over the Program·s lifetime and important operational and financial efficiencies. If a physical loss is detected and repaired, the savings will be in reducing variable operational costs for the Régies as less water needs to be produced and pumped. This would create additional water supply and allow new customers to connect to the supply system. These savings would also reduce energy consumption and GHG emissions, relieve pressure on groundwater, and delay the need to develop alternative water supply sources. When a commercial loss is detected and resolved, the saving will be an immediate revenue increase and thus is based on the water sales tariff, generating additional revenues, and improving the financial performance of the beneficiary service providers. Establishing DMAs also makes optimizing energy consumption, pressure management, and water quality issues possible.

District Metered Areas for more efficient water loss reduction

Water losses are defined as the volume of water distributed minus the volume of water billed in each network. Therefore, water losses encompass physical losses due to leakages in the water distribution system and commercial losses due to faulty meters or illegal connections. When a water network is operated as an open system, determining water losses can only be calculated for the entire network, which is an average for the entire system.

Moving from passive to active leakage control allows for more efficient operations, and it requires the sectioning of the larger network into smaller zones or district meter areas (DMAs). A DMA is defined as a discrete part of a water distribution network. The concept of monitoring flows into DMAs, where bursts and leaks are unreported, is now an internationally accepted and well-established technique to determine where leak location activities should be undertaken.

The NCSP establishes three main objectives: (a) ensure that all relevant stakeholders, implementing agencies, and partners are fully informed about the pillars of the PNAEPI; (b) promote the participation and engagement of all relevant stakeholders in the implementation of the PNAEPI; and (c) raise the awareness of the general population on the country·s water scarcity situation and the need to change behavior towards water conservation. The Program will support the implementation of the actions related to the NCSP·s third objective due to its strategic role in boosting a water savings culture in Morocco. Despite the country·s water scarcity, many water users are often disengaged and uninformed of the objectives of the PNAEPI, are not aware of the actions they can take to conserve potable water and water for irrigation and may view non-conventional water resources with mistrust.[7] 3 These issues may hinder the successful implementation of the Program and the PNAEPI.

The Program will support the roll-out of communication campaigns to galvanize support for implementing the PNAEPI and incentivize awareness of water consumption reduction among the general population.

These campaigns seek to engage with the general population on the main pillars of the PNAEPI, raise awareness about the water scarcity challenges the country faces, encourage support to augment non-conventional water resources in the country·s water mix, and incentivize behavior change towards water conservation. Ultimately, the campaigns also seek to build trust between water users and water management institutions.

Enabling the Integration of Non-Conventional Water Resources

46. A sound regulatory framework and an enabling environment must accompany the successful expansion of non-conventional water resources. Diversifying the water resource matrix is critical for Morocco to boost water security and resilience by decoupling water production from hydro-climatic variability.

The Program will support capacity-building efforts for contracting and managing Public-Private Partnerships (PPPs) for desalination. As indicated by the NDM report and the PNAEPI, given the high-CAPEX costs associated with seawater desalination, mobilizing private-sector financing is key to Meet Morocco·s water security objectives.

PPP contract models allow for the assessment of asset performance (desalinated water production capacity, quality, and water pressure at the point of delivery) against agreed-upon technical specifications, timing, and budget with the private party.
The remuneration to the private developer can be easily linked to the demand for desalinated water using capacity-plus-volume tariff structures. In 2016, 47 percent of desalination capital investments in the world (US$2.9 billion) were financed using PPP schemes, with the MENA region leading with more than 30 desalination BOTs signed in the past 20 years.[7] 5

Wastewater reuse scale-up

48.

In 2006, the government of Morocco embarked on a large National Sanitation Program (Programme National d·Assainissement Liquid et d·Épuration des Eaux Usées, PNA), which resulted in about 84 percent of the Moroccan urban population connected to a sewerage network in 2019 and increased the level of wastewater treatment from 8 percent in 2005 to 53 percent in 2019. By the end of June 2020, Morocco had completed 153 wastewater treatment plants (WWTPs), with an overall treatment capacity of about 3.4 MCM per day. Most of this installed capacity in volume concerns pre-treatment and discharge at sea via an outlet (2.3 MCM per day or 68 percent of the installed capacity) because the large coastal cities (Rabat, Tangier, Casablanca, Tetouan) have retained this type of treatment system. In addition, 16 percent (or 0.5 MCM per day) of the installed capacity concerns tertiary treatment, followed by 13 percent at the secondary stage and 3 percent of the capacity for primary treatments exclusively. These volumes represent an important resource to alleviate the water scarcity challenges the country is facing.

The PNAM, part of the PNAEPI, establishes that further developing and expanding treated wastewater reuse is a strategic solution to Morocco·s water scarcity challenges. The PNAEPI foresees a total investment of MAD 3019 million (US$301 million) to expand treatment capacity and upgrade WWTPs from primary or secondary treatment to tertiary treatment for reuse. The proposed investments build on Morocco·s successful experience implementing reuse projects, for example, for use in industry, green spaces, and golf courses across the country.
In 2020, 21 golf courses were already being irrigated with treated wastewater. The PNAEPI foresees additional 87 reuse projects to be implemented under 2027.

The implementation arrangements of the reuse projects are established in dedicated reuse agreements. These agreements establish the purpose and scope of the project, as well as the roles and responsibilities of the involved

[7] 5 Global Water Intelligence (GWI), 2017.

The average tariff charged for the treated wastewater is between 2 to 3 MAD per m3.

Non-conventional water resources use in industry · the case of the OCP group

Morocco has the largest reserves of phosphates in the world (75 percent of the global stock). The state-owned OCP group is responsible for the exploration, extraction, processing, and marketing of phosphate rock and its derivatives.

As part of its sustainability strategy, OCP has committed to adhering to circular economy principles and is currently meeting around 30 percent of its water demand through the use of treated wastewater and desalinated water.

In line with the PNAEPI and the OCP Group·s sustainability strategy, the OCP Group has committed to cover all its water needs with non-conventional water resources by 2030 (72 with water from desalination plants and 28 percent with treated wastewater, for a total water need of 161 MCM per year by 2030). For this, several projects are underway including the expansion of existing desalination plants and the construction of the Tadla WWTP, and the Fqih Ben Salah WWTP. The existing WWTP at Khouribga will be expanded to treat twice the current volume. The OCP also has in place a large program for increasing the efficiency of its water processes, including recycling 80 percent of the water used in phosphate enrichment process; and saving three MCM of water a year through the installation of a slurry pipeline that has replaced conventional railway transportation that used to transport phosphate from Khouribga to Jorf Lasfar.

By 2021, 87 percent of OCPs energy needs were met by co-generation and wind energy. As part of its Sustainability Strategy, the OCP will expand its solar power grid as well as engage in energy efficiency practices to reduce consumption by 10 percent.

The Program will support the expansion of treated wastewater available for reuse to more than 52 MCM per year by 2027.

The scope of each reuse project will be specified in its corresponding reuse agreement, and it will include: the upgrading of existing wastewater treatment plants from primary or secondary treatment to tertiary treatment, the extension of tertiary treatment capacity, and the construction of treated wastewater distribution network and associated infrastructure (pumping stations, reservoirs, etc.).

The Program will strengthen the legal framework for reusing treated wastewater in agriculture by updating the by-laws regulating its use. The Water Law 36-15 provides bylaws for the safe and sustainable reuse of treated wastewater in various sectors, including agriculture, industry, and public spaces.

It shall encompass dynamic planning that reflects the increasing uncertainty derived from climate change; and the principles to strengthen the sector's governance, institutional, and financial aspects (including defining water valuation and cost-recovery principles). The preparation of the Plan entails significant coordination efforts among the relevant stakeholders, including the MEF, MEE, MoI, MAPMDREF, OCP, and service providers (irrigation and water supply, and sanitation services). It reflects the water needs of sector-specific Plans (notably the Generation Green and OCP Water Plan) Its approval must be done by the Superior Council of Water and Climate, chaired by the Prime Minister.

DLI#2: Groundwater management improved. This DLI seeks to support the GoM efforts to improve groundwater resources and ensure long-term sustainability. This DLI will incentivize the development, adoption, and implementation of legal provisions and consultative processes for participatory aquifer management contracts. Also, the DLI provides incentives for signing participatory management contracts in three significantly overexploited aquifers (Berrechid, Maamoura, and Bahira). Finally, the DLI supports GoM's effort for improved groundwater management by deploying smart meters to monitor groundwater abstraction by large users in aquifers within the Program area.

This DLI provides incentives to strengthen the performance of the river basin agencies (ABHs), key institutions for water resource management at the basin level.

DLI#4: Water Information Systems operationalized and used for decision-making.

To this end, the DLI will incentivize the adoption of the Decree establishing the SNIEAU and ABHs' Water Resources Information Systems and the operationalization of the systems, which entail: (a) interoperability between the SNIEAU and ABHs' Water Resources Information Systems; (b) interoperability with relevant databases (stored water volumes in dams, inventory of wells by the National Conservation Agency, water quantity and quality, among others); and (c) their accessibility to the public (partial). Finally, the DLI provides incentives for publishing in the SNIEAU of the annual hydrological status by selected ABHs included in the Program, including uses and quality aspects.

This DLI aims at establishing the foundations of the regulatory systems for the SRMs- including two major elements: (a) KPI reporting requirements feeding into an MoI digitalized performance information system; and (b) the definition of minimum service standards for electricity and water supply distribution and the wastewater collection and treatment.

This DLI seeks to improve the sector's financial sustainability across the entire water value chain by developing a Financing sustainability framework comprising a financial model et financial sustainability action plan.

In particular, the DLI will support developing a financial model, and adopting a financial sustainability action plan.

DLI#7: Volume of potable water savings in distribution water supply networks.

Through the Program, 20 million m3 of potable water will be saved by Régies in the Program areas through network sectorization and pressure management, improved leakage control and repair, deployment of GIS tools, and rehabilitation of distribution mains.

Through this DLI, the Program will support the revision of the bylaw defining the norms for wastewater reuse for irrigation purposes, noting that the general use of wastewater is already adopted.

The Program will make 52 million m3 of treated wastewater available for reuse (equivalent to 52 percent of the PNAEPI target).

Preparation of participatory aquifer management contracts.

The Program will incentivize the development of water information systems and capacities to improve the quality of water data, with a particular focus on user accessibility.

The PNAEPI communication strategy supported by the Program aims to build public support for the strategic priorities of the PNAEPI and promote sustainable water management practices.

These will be once again evaluated at the end of the Program to assess improvement in the suitability of messages, actions towards water conservation, and support to the Program activities.

Establishing DMAs for non-revenue water management will allow the Régies to respond more quickly to customer complaints about low water pressure or other issues related to the distribution network, as leakages can be more readily identified and fixed. Régies will also organize public meetings and community outreach campaigns during the implementation of non-revenue water reduction actions.

Rising temperatures and erratic rainfall have reduced river flows and increased evaporation and siltation of storage dams, leading to a 20 percent reduction in overall water resources in the last 30 years, exacerbating the problem of water scarcity generated by non-climate stressors such as population growth in the north, irrigation expansion, as well as urban, industrial and tourism development.

Reducing energy consumption resulting from reduced water losses will generate GHG emissions savings by reducing the overall energy usage by the water sector.

The NPV of these emissions is US$3 million.

Synthesis of the relevant expenditures associated with the Program (2023-2028)		
Implementing Agency	Million MAD	Million US$
ABH	1,822	180
MEE	487	48
Regies	2,012	199
Reuse projects	1,480	146
Total	5,801	573

Source: Annual budgets and concerned entities' budgets and forecasts

Budget structure and lines

79. The PNAEPI is funded through the budget of the MEE, the ABHs included in the Program (Loukkos, Sebou, Bouregreg Chaouia, Oum Er Rbia, Tensift and Souss Massa), and the budgets of the 11 Régies included in the Program (RADEEF, RADEEJ, RADEEL, RADEEMA, RADEES, RADEET, RAK, RAMSA, RADEETA, RADEM and

At the central level of the MEE, the investment budget is under Program 601, entitled "Water," which includes several projects organized by budget lines. These lines encompass studies, procurement, and works on evaluating surface and groundwater water resources and management and IT tools, platforms, and solutions development.

The expenditure framework encompasses the budget lines that support the development of water resources through the water public domain, water resources evaluation, planning and management, and water resources preservation and control.

RBA Budget lines considered for the Expenditure Framework

Budget line	Title
10	Modernization of the administration
20	Studies and research
30	Management of the Hydraulic Public Domain
40	Water resources evaluation
50	Water resources management and planning
60	Water resources conservation, preservation, and protection
80	Maintenance of Hydraulic Public Domain Assets
40 (Fct)	Communication and awareness

82.

These lines cover the tertiary treatment of 36 communes, representing 11 percent of the projects considered under the National Mutualized Sanitation Program (PNAM).

Regarding the 11 Régies included under the Program, the expenditure framework covered all the budget lines that support the implementation of the RA2 related to water savings in distribution water supply networks. The analysis identified the following activities under the budget lines of the 11 Régies: (a) renewal and reinforcement of the distribution network; (b) leaks detection campaigns; (c) pressure management and network sectorization; (d) acquisition of leak detection equipment; ed) renewal of water meters and installation of electromagnetic flowmeters for large consumers; and, (f) GIS deployment.

The economic, social, and financial measures taken rapidly by the government to offset the negative economic impact of the crisis have contributed to reducing the budgetary and financial margin of the government.

The 2023 Finance Law provides for the continued construction of the ·Social State· and the strengthening of its pillars via concrete and unprecedented measures that will improve the living conditions of large segments of the population, in particular, the generalization of social protection to vulnerable and needy categories, the improvement of the purchasing power of the middle class and the healthcare offer, the promotion of public schools and the development of mechanisms for access to housing.

Also, among the four general orientations of the 2023 Finance Law, one of the four axes retained in this project concerns ·the restoration of the budgetary margins to ensure the sustainability of the reforms. The Government will attach particular importance to financing large-scale structural reforms initiated in recent years, to mobilize more resources and ensure their sustainability.

Among the sectoral strategies announced, the 2023 Finance Law provides within the framework of the PNAEPI "the implementation of urgent measures to ensure the supply of drinking water, through the launch of a multitude of projects related to the transfer of water between certain basins, the implementation of seawater desalination stations as well as the reuse of treated wastewater in the watering of green spaces, industrial uses and the needs of tourist complexes."

If the tools for defining an overall budget policy strategy are of acceptable quality at the level of the entire budget, they are still under construction at the level of the ministerial sectors.

The quality of the monitoring of budget execution during the year, despite the delays noted, also promotes the proper allocation of resources through the possible rapid readjustments that it allows.

The cost-benefit analysis estimates the economic feasibility of the Program by calculating the net present value (NPV) of cost and benefit streams and by determining the Program's economic rate of return (ERR).

The cost-benefit analysis included economic costs and benefits under RA2 and RA3. It was undertaken from an economic perspective converting financial cash flows into economic cash flows to eliminate distortions caused by taxes, subsidies, and other externalities. Costs include the investment and O&M costs of schemes under RA 2 and RA3 during the Program life.

The net benefit is the difference between the incremental benefits and the incremental costs of two scenarios: ·without· and ·with· the Program.

Results under RA 2 and RA 3 were evaluated, measuring the expected costs and benefits for the Program·s lifetime, estimated at 20 years for the investments included under RA2 and RA3.

· Costs and benefits are expressed in constant prices as of 2023.

The analysis was completed using a 6 percent discount rate, corresponding to the Bank directives regarding discount rates for economic analysis.[81]

The expected economic benefits of these investments include (a) increased water revenues resulting from physical and commercial loss reductions; (b) decreased operational costs as a result of leak detection and reduced energy requirements; (c) avoided capital expenditures; (d) GHG emission reductions owing to reduced energy requirements.

Additional water sales arising from greater water availability derived from reduced water losses. Reducing water losses is a key priority of the PNAEPI to help address water scarcity in Morocco and generate financial returns for the operators. The rehabilitation of the distribution network under the Program aims to reduce commercial and physical losses across 11 Régies, which will reduce bulk water purchased from ONEE. As of 2023, the 11 Régies in the Program purchase bulk water from ONEE to bridge supply gaps. Investments in non-revenue water (NRW) reductions will increase the water sold directly by the 11 Régies and reduce the bulk water purchases from ONEE. As a result of the project, 20,000,000 m3 of water savings are expected to be achieved over the project's life with an average sale price of 4 MAD/m3 (US$ 0.61/m3),[82] generating additional revenues of up to US$8 million per year. Investing in NRW reductions has been shown to reduce future capital expenditures owing to the extension of the useful life of current assets and a reduction in the rate of asset replacement.

Incremental O&M Savings. The economic analysis considers incremental O&M savings from reducing production costs due to reduced pumping costs resulting in reduced energy consumption. Reducing water losses will translate into a reduction of 1, 77 MWh over the Program·s life, leading to O&M savings of US$7.

Water loss reduction programs have been shown to reduce the need to replace capital equipment in the network by extending their useful life. Water loss reductions are expected to reach 54,000 m3/day, leading to savings of US$55 million in capital expenditure.

Discounting Costs and Benefits in Economic Analysis of World Bank Projects.

[82] Calculated using a weighted average of water sold and sale price in 2022 for 10 of the 11 Régies as data was unavailable for RADEETA.

Reducing energy consumption due to reduced water losses will generate GHG emissions savings by reducing the overall energy usage by the water sector.

Investment costs of US$ 192 million include capital expenditures under RA2 related to the implementation of water loss reduction plans, including the deployment of geographical information management systems and hydraulic models; meters deployment (bulk- and micro-meters), network sectorization and pressure control Program; leakage detection and rehabilitation campaigns. O&M costs (estimated at US1.4 million annually) are expected to increase for NRW reduction. They can be easily offset by reductions in pumping costs and capital expenditure savings due to water loss reductions.

RA3 investments for increased wastewater reuse volumes. The expected economic benefits include (a) GHG emission reductions owing to a reduction in local pollutants from WWTP effluents; and (b) revenue from the sale of treated wastewater.

WWTP pollutant reductions. Program benefits stemming from the reduction in local WWTP pollutant loadings are estimated using a shadow price of pollutants for Biological Oxygen Demand (BOD), Chemical Oxygen Demand (COD), Total Suspended Solids (TSS), and Nitrogen (N).[83] The reduction in these pollutants arises from upgrades to WWTPs to increase the capacity for tertiary treatment. As a result of the Program WWTPs will be upgraded from primary or secondary treatment capabilities to tertiary treatment resulting in large local pollutant reductions of BOD, COD, TSS, and N. The avoided costs resulting from removing pollutants during wastewater treatment, known as the shadow price of all pollutants, were used to estimate the economic value of these pollutants. These values depend on the receiving body of water, as reducing these pollutants affects the local biodiversity.

Upgrading WWTPs to include tertiary treatment will allow TWW to meet the acceptable standard for reuse and will dramatically increase the amount of TWW sold. Expanding the use of treated wastewater will also help address water scarcity by increasing water availability.

Program costs Treated Wastewater Reuse.

Economic valuation of wastewater: the cost of action and the cost of no action.

Besides direct economic benefits, many other potential benefits are not factored into the cost-benefit analysis. This is either because estimating such benefits is difficult due to the lack of data or it is challenging to quantify the value of those benefits because they might not be financial or economic.

Supporting the enabling environment and regulatory framework that facilitates the scaling up of non-conventional water resources provides Morocco with a viable path to securing its water future.

Development of a water valuation strategy. The valuation of water is key to informing water pricing and allocation decisions and enables the most efficient allocation of an increasingly scarce resource.

Economic Rate of Return (ERR) and Net Present Value (NPV)

Scenario	ERR (%)	NPV @ 6% (US$ million)
Without shadow price of carbon	15.5	230
With the shadow price of carbon	15.5	231.8

Source: Team Calculations, June 2023.

	Strengthened Water Sector Governance	2,309	228	40%
RA2	Improved Financial Sustainability and Water Use Efficiency	2,012	199	35%
RA3	Enabled Integration of Non-Conventional Water Resources	1,480	146	26%
Total PforR		5,801	573	100%

5. Except for the MoI, which is involved in several Bank-financed operations, the remaining participating implementing agencies have no experience implementing Bank-funded operations (PforR). Overall fiduciary performance of PforR operations managed by the MoI is deemed acceptable.

Also, with the implementation of recommended mitigation measures, before or during the program implementation, the capacity and performance of the remaining implementing agencies will be adequate to provide reasonable assurance that the Program funds will be used for the intended purposes.

The audit reports will be submitted to the World Bank within nine (09) months following the end of each calendar year.

The overall residual fiduciary and procurement risk of the Program is assessed as Substantial. The IFSA concludes that only upon implementing the agreed fiduciary mitigation measures will the Program·s fiduciary systems provide reasonable assurance that the financing proceeds will be used for intended purposes. The following key fiduciary risks were identified: (i) weaknesses in capacity for procurement planning and execution, (ii) challenges in operationalizing the new procurement decree, (iii) an inadequate procurement complaint handling system, (iv) a lack of procurement performance reporting mechanisms and (v) a lack of suspension and debarment check mechanism which might result in awarding a contract to firms and/or individuals debarred or suspended by the Bank, (vi) low budget commitment and payment by some ABHs and Regies, (vii) delayed preparation of consolidated financial statements and submission of program audit reports due to the large number of implementing entities located around the country which requires an effective and adequate fiduciary coordination mechanism, (viii) the lack of familiarity with PforR financing instruments of most of the entities (e.g., ABHs, Regies and DGH) involved in the implementation of the program combined with the use of spreadsheet ·Excel· to prepare consolidated financial reports may impact the fiduciary performance of the Program, including the quality and timeliness, and (ix) there are also some challenges in the follow up of audit recommendations and the internal audit unit at the MEE as per the new decree that needs to be implemented.

III. Legal and Institutional Public Finance Management (PFM) Framework in Morocco applicable to the Program

10. The legal and institutional Public Finance Management (PFM) framework is deemed acceptable for the Program.

Overall, the planning and budgeting of all central entities involved in the Program follow a structured, timely, and disciplined process consistent with the country·s PFM cycle and ensures that allocations fit within the available budget envelope.

The 2023 Public Expenditure and Financial Accountability (PEFA) assessment will provide some updates on the performance of the country·s PFM systems applied by the DGH-MEE.

[8] 7As to RA3, the selection

[8] 5 Article 5 describes that the "the annual budget law is drawn up by reference to a three-year program updated each year in order to adapt it to the country's changing financial, economic and social situation.

[8] 6 On December 29, a new procurement decree, decree# 2.22.431, was adopted by the Moroccan government in response to the recommendations of the special commission for the New Development Model.

[8] 7 The new decree aims to consolidate the public procurement system to provide more clarity to economic actors and improve the business climate.

The overall costs of the supported Program, including the portion financed by the World Bank, are expected to be integrated into the budget laws 2024-2028.
Based on budget instructions/circulars, the teams of the various implementing entities under the responsibility of the Directorate of Financial Affairs (DAF) of each participating entity will prepare the Program budget considering the limits of allocations set by the GoM and the Program. The Program·s consolidated budget will be integrated into the budget of line ministries and implementing entities.

The Bank fiduciary and task teams will closely monitor this strategy·s budget programming year by year with the MEF and ministries covered by the Program to avoid incoherence between the Program expenses and the disbursement rate. No specific issue related to funding predictability was identified in Morocco·s ongoing PforR operations.

Evolution of the Budget Allocation of the MEE (MMAD)

through the establishment of a procurement observatory and to promote sustainable development.

The PNAEPI is registered under the budget code 1222017000 with the Expenditure nomenclature heading ·Investment Chapter (MEE).

(ii): Regies and ABHs: They receive subsidies to contribute to financing certain projects included in the programs resulting from the policies for developing the water management and liquid sanitation sectors. This is done within the framework of multiparty agreements established following the requests for subsidies addressed to the two Ministries supported by the details of the investment programs planned by the entities and the related financial arrangements. The draft budgets are prepared by the Authority and presented to the DRPL of the Ministry of the Interior (Regies) and DGH-MEE (ABHs) for approval of the investment projects in terms of consistency with sector-wide policies, strategies, and programs for the management of water, electricity, and liquid sanitation services. Then this same draft budget is presented to the Directorate of Public Enterprises and Establishments (DEPP) under the Ministry of Economy and Finance during the budget discussion meeting to approve the projects, the commitment and payment appropriations entered, as well as the planned financing methods to ensure the financial balance of the proposed draft budget.

Based on the activities identified in the Program scope, the main procurable items include:

(a) Works: water leakage detection and rehabilitation; works to manage, protect, and preserve the hydraulic public domain and aquatic environments; upgrading wastewater treatment plants (WWTPs); and distribution systems for the conveyance of treated wastewater reuse, including pipelines, pump stations, and storage tanks.

(b) Goods and non-consulting services: network sectorization; water data management and information management systems, including upgrade, equipment, and maintenance; equipment, measurement materials, and supplies for water resources monitoring and evaluation and for water losses detection; smart meters; communication campaigns to raise awareness of the importance of water conservation; machinery and equipment maintenance; equipment and materials for hydro-climatological and groundwater measurements; and computer-assisted maintenance management.

(c) Consultant services: evaluation of the institutional roles and responsibilities in key processes in the water sector; the development, adoption, and implementation of regulatory instruments and consultative processes to improve implementation of participative aquifer management contracts; development, implementation and adoption of a performance benchmarking framework to strengthen the performance of selected ABHs to deliver on their core functions of planning, managing, developing and protecting water resources; and the definition and implementation of water losses reduction plans.

The implementing units shall report to the World Bank if large contracts appear throughout project implementation. In addition, the World Bank team will analyze and monitor the Program performance of fiduciary systems and contract management reports to identify any large-value contracts that may appear throughout the Program implementation.

The level of execution of the MEE·s overall budget is considered satisfactory and is above 95 percent over the three years, with a payment rate of around 70 percent.

However, the overall payment rate is around 50 percent, except for ABH Loukkos and Oum-Er-Rbia, which are 61 percent. Concerning the investment budget, the overall execution rate of the ABHs is well above 90%, apart from ABH Oum-Er-Rbia, which has an average rate of 70%.

Hence, attention should be paid to the capacities of ABH Souss Massa and Our-Er-Rbia to execute investment budgets and ABH Sebou and Bouregreg et Chaouia for capital expenditure payments.

Regies level of execution of operating budgets and payment rates of the budget executed is overall satisfactory.

The Program·s funds will be reflected in the Government budget under the MEE (DRPE and ABHs) and MoI (selected Regies).

Specifically, the funds will flow directly from the TSA to service providers, consultants, and constructors to pay invoices for the activities implemented by the participating ministries and entities.

All these measures have significantly improved over the last five years in the average payment time of EEP.

The funds will be disbursed to the government·s Treasury Single Account (STA) for advances, prior and achieved results.

If the World Bank finds that the disbursement request meets the Financing Agreement terms, the World Bank will disburse the corresponding funds to the Treasury Bank Account opened at the Central Bank (Bank Al-Maghrib).

The Government, through its budget execution procedures, will transfer its contribution to the Program into the Treasury bank account managed by the public accountants assigned to the implementing entities participating in the Program.

Participating entities from the MEE will apply similar accounting standards for the Program using the Integrated Expenditure Management (IEM) system (GID), an integrated set of computerized applications developed in-house.

The format of the table summarizing the semi-annual financial information (execution of the PEF, amounts committed and paid) and annual financial statements will be defined in the MOP and presented to the FM teams of the implementing entities during the Program launch and dissemination workshop of the MOP. The DRPE will centralize financial information and oversee the work of the PCU, including financial reporting. ThePNAEPI·s Technical Committee will oversee Program activities and management, including fiduciary work.

The annual financial statements will include the financial execution of each Result Area, and the data will be collected from participating entities. The head of the financial management team of each ABHs and Regies will be responsible for preparing and submitting to the PCU the financial reports on the budget execution of their activities implemented.

Specifically, similar to most of the Bank-financed operations in Morocco, a spreadsheet ·Excel· and data extracted from the national budget execution software ·Integrated Expenditure Management System (Gestion Intégrée de la Dépense-GID) and Integrated Revenue Management System (Gestion Intégrée de la Recette- GIR), will be used to prepare Program·s periodic consolidated budget execution reports and consolidated annual financial statements which in turn, may impact the quality and timeliness in the preparation of those reports and submission of audited financial statements.

Implementing agencies· bidding documents is based on the guidance note and the standard bidding documents issued by TGR in 2015.

Procurement under Wastewater reuse projects

37. 17 wastewater treatment projects for $146 million are identified as part of the Program scope and included in the PNAEP Municipalities are responsible for ensuring that wastewater is collected and treated through direct service provision of delegated management.

Thus, ONEE acts as a delegated project management entity for some infrastructure projects and manages some services for municipalities or the State.

Additionally, with the generalization of electronic submission, contracting authorities must establish an electronic procurement record management system that complies with the technical and legal requirements established by the Ministry in charge of finance. While some implementing agencies did not provide details about how record keeping is implemented, this re uirement is generally complied with, given Morocco·s robust audit institutions and legal framework.

The assessment looked at data available on the MEF platform AJAL to assess the performance of contracts payment.

Furthermore, the TGR has implemented the Moroccan Public Procurement Portal (e-Government Procurement) and integrated systems to track and implement budget spending and process payments (GID).

A recent assessment by the World Bank of the Moroccan e-GP system concluded that the

[9] 2 This platform is intended for suppliers of EEP members to electronically submit invoices and/or claims for late payments. The purpose of this platform is to: (i) Provide a space for suppliers to submit and process invoices and/or claims regarding payment delays by EEP members; (ii) Facilitate communication between suppliers and EEP members; (iii) Identify the causes of payment delays and provide appropriate solutions; (iv) Improve payment deadlines for suppliers of EEP members; (v) Digitize invoice and claim operations related to payment delays; (vi) Strengthen transparency and traceability in the processing of invoices and claims; and (vii)Hold stakeholders accountable and control the processing time for invoices and claims by EEP members.

One representative from each implementing agency shall be nominated for granting access to the Client Connection by the World Bank. The borrower will develop and operationalize the mechanism of enforcing these requirements through the issue of instructions/circular to all the procuring entities requiring the procuring officers to check the eligibility of firms and individuals from the Bank·s list of debarred and suspended firms and record the same in procurement award decision files.

The terms of reference setting out the modalities of intervention of these institutions will include specific provisions relating to verifying compliance with the Bank·s guidelines on preventing fraud and corruption.

The audit reports and detailed management letters will be submitted to the Bank no later than nine (9) months after the closure of accounts.

Fiduciary Management Capacity Assessment of Implementing Agencies.

Overall, except the MoI (Regies excluded), the assessment of the capacity of the fiduciary staff of the entities involved in the Program concluded to the need to implement a capacity program.

This PCU will include a fiduciary officer ensuring cooperation and consolidation of The Program·s financial and procurement information prepared by each PCU.

Financial Management: The Public Accountant of the entities involved in the Program will execute the budget following the public expenditure chain through the budget execution software GID and GIR.

Delegatees in charge of wastewater projects under RA#3 shall abide by this requirement.

The aim is to ensure that the Program does not involve significant E&S risks and that the systems put in place make it possible to identify and manage any risks. In particular, the ESSA identifies and analyzes the gaps between the national systems and the principles applying to the Program and recommends actions for improvement aimed at the consistency of the E&S management systems with the requirements of the policies and Bank guidelines applicable to the Program.

Two categories of activities will be eligible for funding under the Program: (a) those related to planning, improving knowledge, or strengthening systems and skills (non-structural activities); and (b) those associated with the reduction of water losses in distribution networks, upgrading of existing WWTPs and associated infrastructure for reuse (small or medium-scale structural activities).

The MEE will manage the Program through a Program Coordination Unit (PCU), which will be the focal point for the World Bank.

The POM will define the staffing requirements for the PCU core team.

Most institutions involved in this Program1 [m1] 3 have experience working with the World Bank in the context of previous development projects and programs in Morocco since the 2010s, except for the River Basin Agencies (Agence du Bassin Hydraulique, ABHs) and the Regies. These include the Blue Economy PforR, the Municipal Performance PforR, the Improving Early Childhood Development Outcomes PforR, the Natural Disaster Management Program, etc. The MEF is involved in all the PforRs financed by the Bank.

A two-step filter was applied concerning E&S risks, following the Bank·s PforR policy approved by the Board of Directors: ·Activities deemed likely to have significant, diverse or unprecedented adverse impacts on the environment and affected people are not eligible for PforR financing and are excluded from the PforR Program.· First, the list of 49 activities proposed for Program funding was analyzed against the exclusion criteria of paragraph 14 of the Bank Guidance Program-for-Results Financing E&S Systems Assessment Based on this review, 27 investments were assessed as highly risky due to their potentially significant negative E&S impact despite potential mitigation measures, and therefore they were excluded from the Program. Second, the remaining 22 investments· potential negative impacts were deemed reversible and reviewed (see Table below) to identify specific mitigation measures.

The tertiary treatment system will recharge wastewater for irrigation and watering green spaces.

and site safety, tertiary treatment stations for liquid effluents, etc.)

Management Plans (ESMPs) for these activities, which will be prepared by the project leaders before the start of the works and which will become binding for construction companies.

strengthening the territories· resilience to climate change, and safeguarding its water resources through more efficient use and management of its scarcity.

The Program is aligned with the World Bank Group·s Country Partnership Framework for Morocco for fiscal years 2019-24.

More specifically, the Program supports intervention area 3, ·Promoting inclusive and resilient territorial development,· directly contributing to the following three objectives: (a) improving the performance of key infrastructure provision services to cities, (b) improving access to sustainable water resources, and (c) strengthen adaptation to climate change and resilience to natural disasters.

With a strong impact on the governance of water resources, the PROGRAM will contribute to promoting a sustainable management model for this resource.

shown the absence/weakness of E&S management systems.

The MEF is familiar with the Bank·s instruments, including the PforR instrument, and has experience managing technical assistance projects.

The DDD is responsible for managing the EIA system and has good experience and the necessary skills, particularly around EIA review, project implementation monitoring, and environmental monitoring (air, water, soil), through the National Environment Laboratory.

The social risks likely to result from the Program are linked during the preparatory phase to potential shortcomings in assessing social impacts and planning for their management and restrictions on access to natural resources.

They will be low-harm, controllable, and manageable based on well-recognized E&S mitigation measures.

To this end, the Program will support specific measures to strengthen the quality and performance of the E&S management systems in two areas: (a) strengthening the E&S management system and (b) building the capacities of stakeholders in E&S management.

Actions aiming at strengthening the E&S management system

- The development of diagnostic tools and E&S monitoring of sub-projects (screening against the negative list, E&S monitoring sheets, ESMP, RAP) and the tools for site monitoring (E&S monitoring sheets, anomalies sheets), which will be included in the POM
- The designation of an E&S focal point in the PCU (trained in E&S management) to ensure the implementation of E&S actions; the collection of information concerning E&S risks; the M&E of the implementation of mitigation measures; and the integration of data in the information system and reporting.

Actions aiming at building stakeholders· capacities in E&S management

- Based on the E&S technical manual, the PCU·s E&S focal point will develop a training plan for all Program stakeholders to provide stakeholders with principles and methods for developing and implementing an E&S management system

- The E&S staff in implementing agencies will ensure -in close collaboration with the PCU·s E&S focal point- the screening against the negative list of the sub-projects, identification of E&S risks and their mitigation measures, and the M&E of their implementation and reporting, including any incidents/accidents that occurred during the implementation of Program activities.

90 days after the declaration of effectiveness, the PCU will carry out -with the support of the Bank·s E&S team- an evaluation of the existing GRMs in the ABHs and the Régies included in the Program to verify the existence of complaint reception, processing, and handling systems following the Decree 2-17-265, and if in absence support the creation of complain management system with an adequate complaints management procedure

- Works contracts.

The Implementation Support Plan is based on the World Bank Guidelines for PforR Operations.

The focus of the World Bank implementation support under the Program will be to:

- Review implementation progress and achievement of Program results and DLIs
- Provide support for resolving emerging Program implementation challenges
- Provide technical support for the implementation of the PAP, the achievement of DLIs and other results, and institutional development and capacity building
- Monitor systems· performance to ensure their continuing ade uacy through Program monitoring reports, audit reports, and field visits; and
- Monitor changes in risks to the Program and compliance with legal agreements and, as needed, the PAP
- Mobilize technical assistance funds to support the achievement of the Program results, as presented below.

A World Bank team will be mobilized to deliver Bank support as mentioned above (see Table A7.2).

After effectiveness, the first implementation support mission will occur as soon as possible to provide direct feedback and implementation support with critical inputs and involvement of environmental, social, and fiduciary teams. The World Bank will support implementation through regular missions and guide implementation.

Finally, the World Bank will mobilize trust funds (Bank or client executed) to support the Borrower in the implementation of the Program. During preparation, some indicative areas of Bank support through trust funds were identified · as indicated in the Table below.

International experience in the development and monitoring of the process of data certification for hydrological and hydrogeological data, and capacity building.